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Wing Box FEA Structural Analysis

Introduction & Methodology Adjustment:

This report details the finite element analysis (FEA) of an aerospace wing box to evaluate stress distribution, deformation, mesh sensitivity, material performance, and modal frequencies. The aerodynamic loads were calculated based on a unique dynamic pressure using the assigned roll number.

During the initial setup, the theoretical minimum thickness formula provided an estimated required thickness of 11.949 m. However, upon testing this thickness under the maximum Wing-Up load case in ANSYS APDL, the maximum stress reached .988E+09 Pa (988 MPa), which catastrophically exceeded the 345 MPa yield strength of the 2024-T3 Aluminium alloy.

Therefore, to ensure no material yielding occurred as per the design requirements, the structural thickness was iteratively optimized and increased to a final safe working thickness of 31.5 m. This optimized thickness is used as the baseline geometry for the analysis.

Data:

Dynamic Pressure: 31184.5 KPa

Lift Force (Up): 1491779.51 KN (Highest force to be used in calculating thickness)

Drag Force (Up): 88796.399 KN

Lift Force (Down): -532778.39 KN

Drag Force (Down): 35518.56 KN

Area: 56.95 m² Chord: 4.87 m

Span: 10.92 m

Thickness for 2024: 11.949 m (Calculated)

Optimum Thickness: 31.5 m

Section 1: Initial Wing Design:

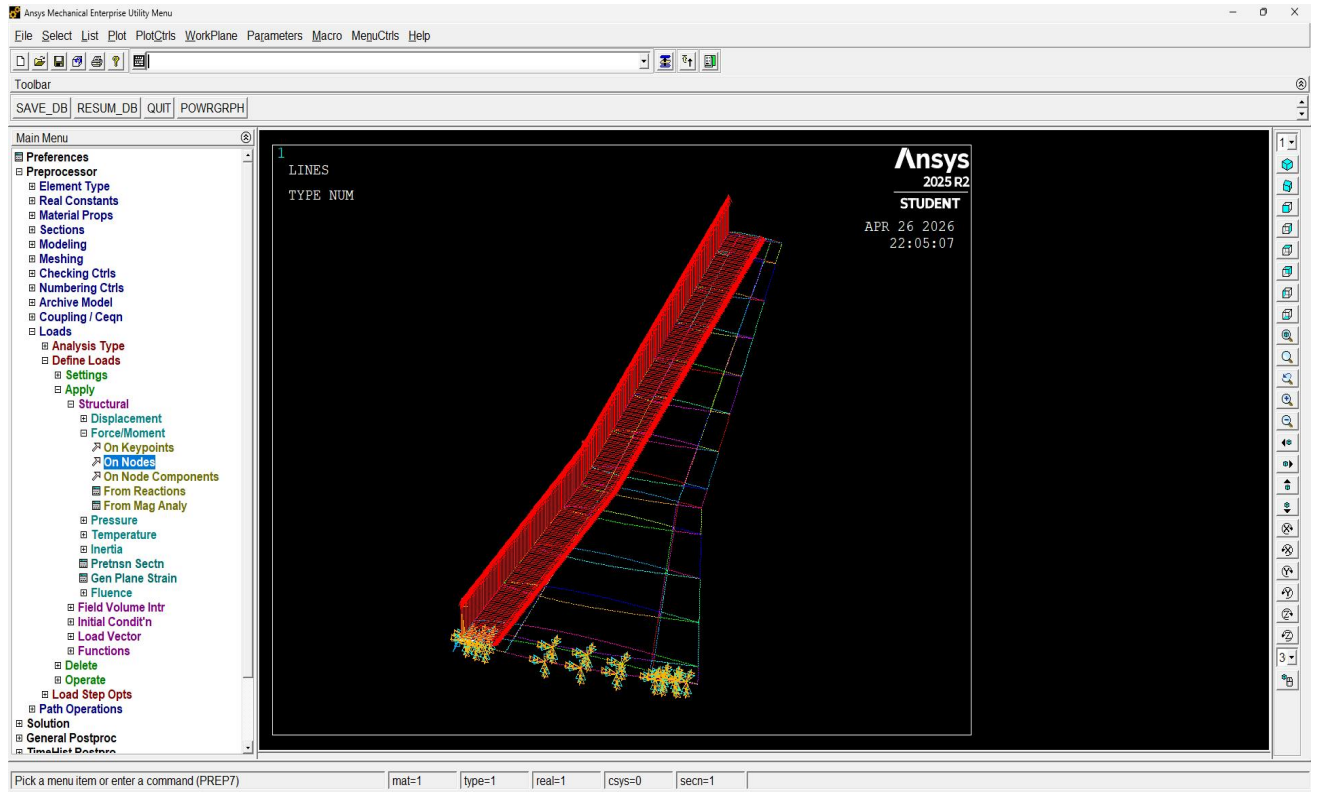
1.1 Wing Geometry: The wing box geometry was imported via IGES format and modelled using linear Shell181 elements. The span of the wing is 10.92 m and the chord is 4.87 m.

1.2 Boundary Condition Analysis: To understand how mounting constraints affect structural integrity, the wing was subjected to two boundary conditions: a standard Cantilever setup (fixed at the root) and a Fixed Ends setup (fixed at both the root and the tip).

Wing-Up Load Case:

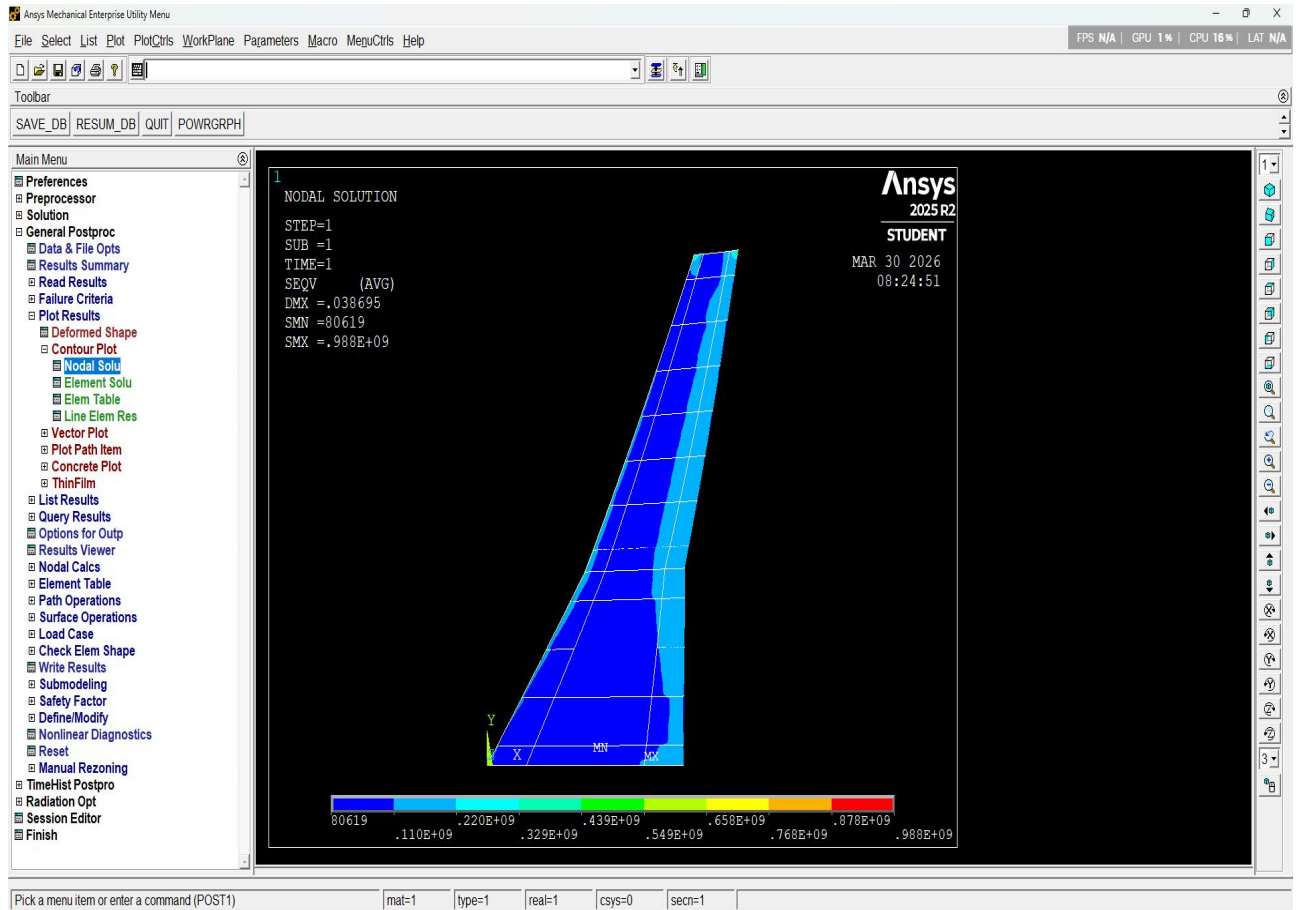
Cantilever: The maximum stress was .176E+09 Pa with a maximum displacement of 0.002238 m.

Fixed Ends: The maximum stress increased to .222E+09 Pa with a maximum displacement of 0.003134 m.

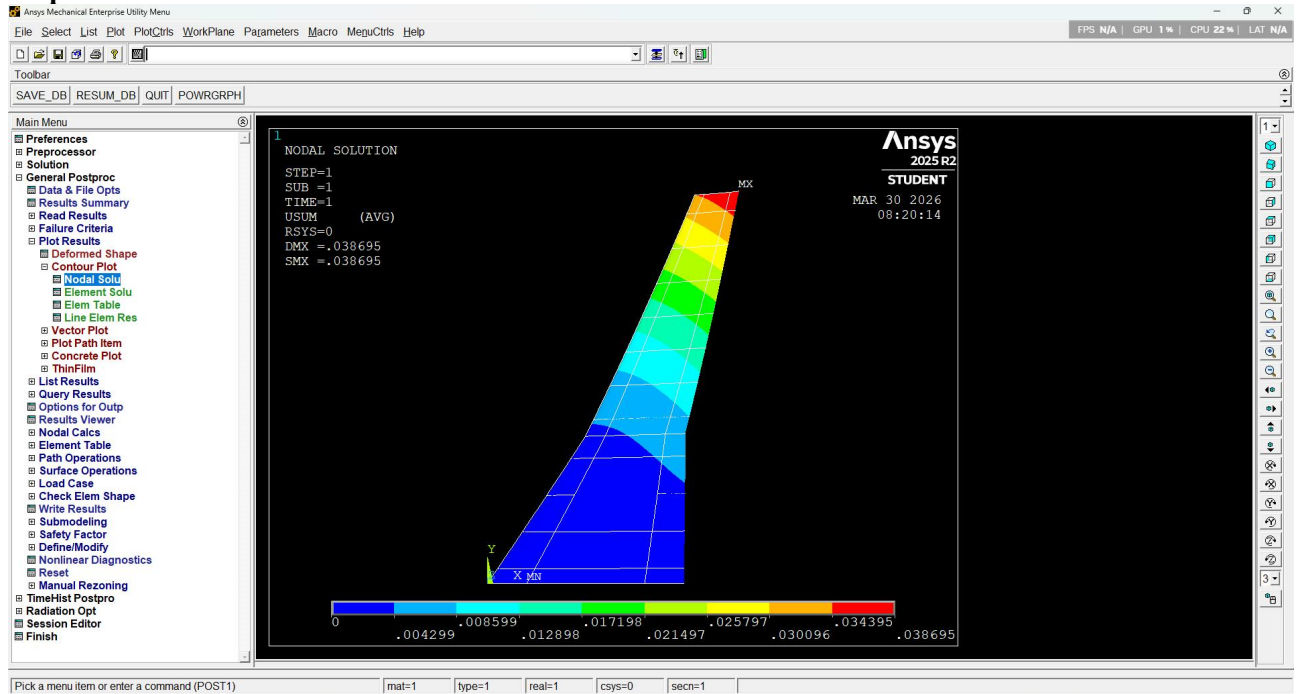


For Cantilever:
(At 11.949 thickness)

Stress:

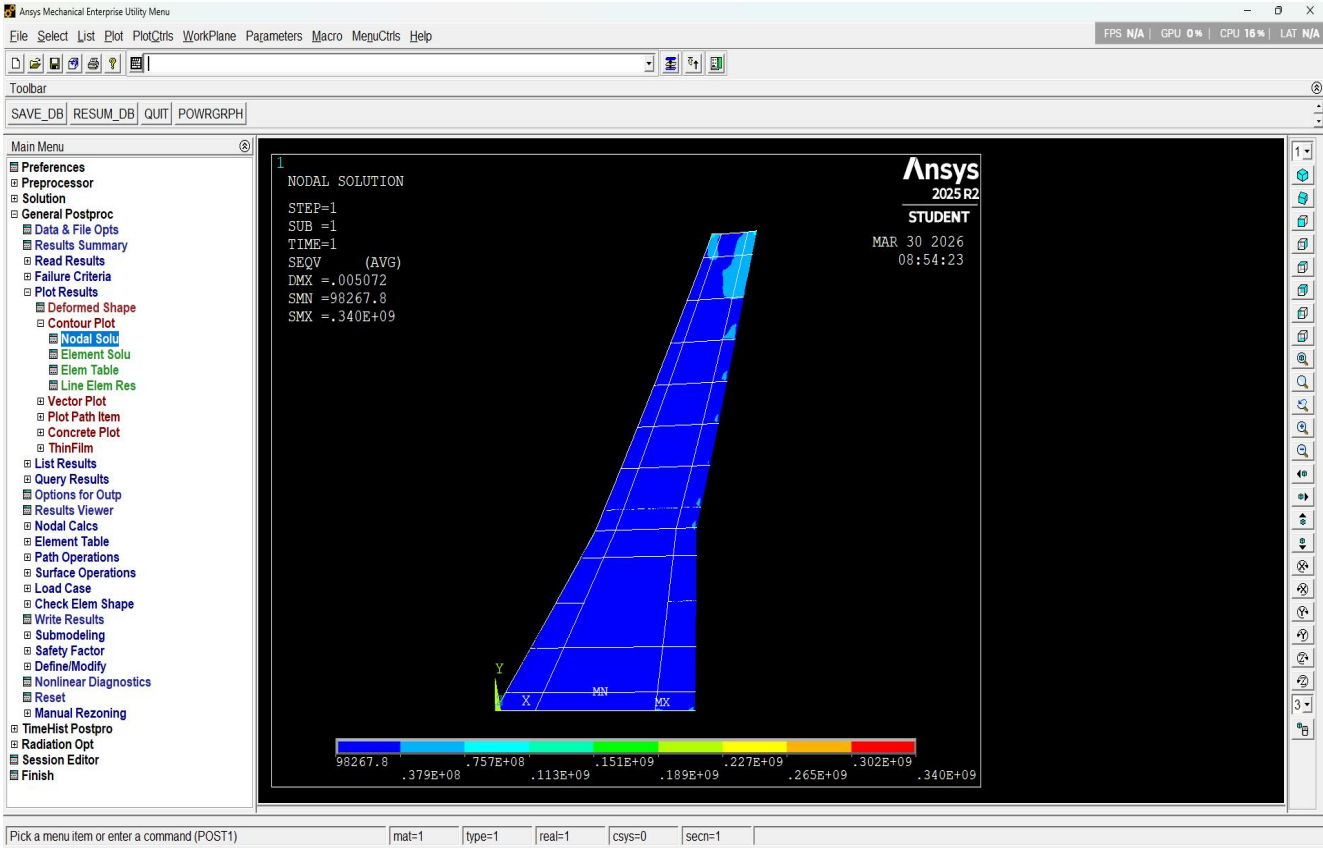


Displacement:



Since at 11.949m thickness the material yields. After testing 31.5m thickness was found to be optimum. All calculations and analysis will use 31.5m as new thickness.

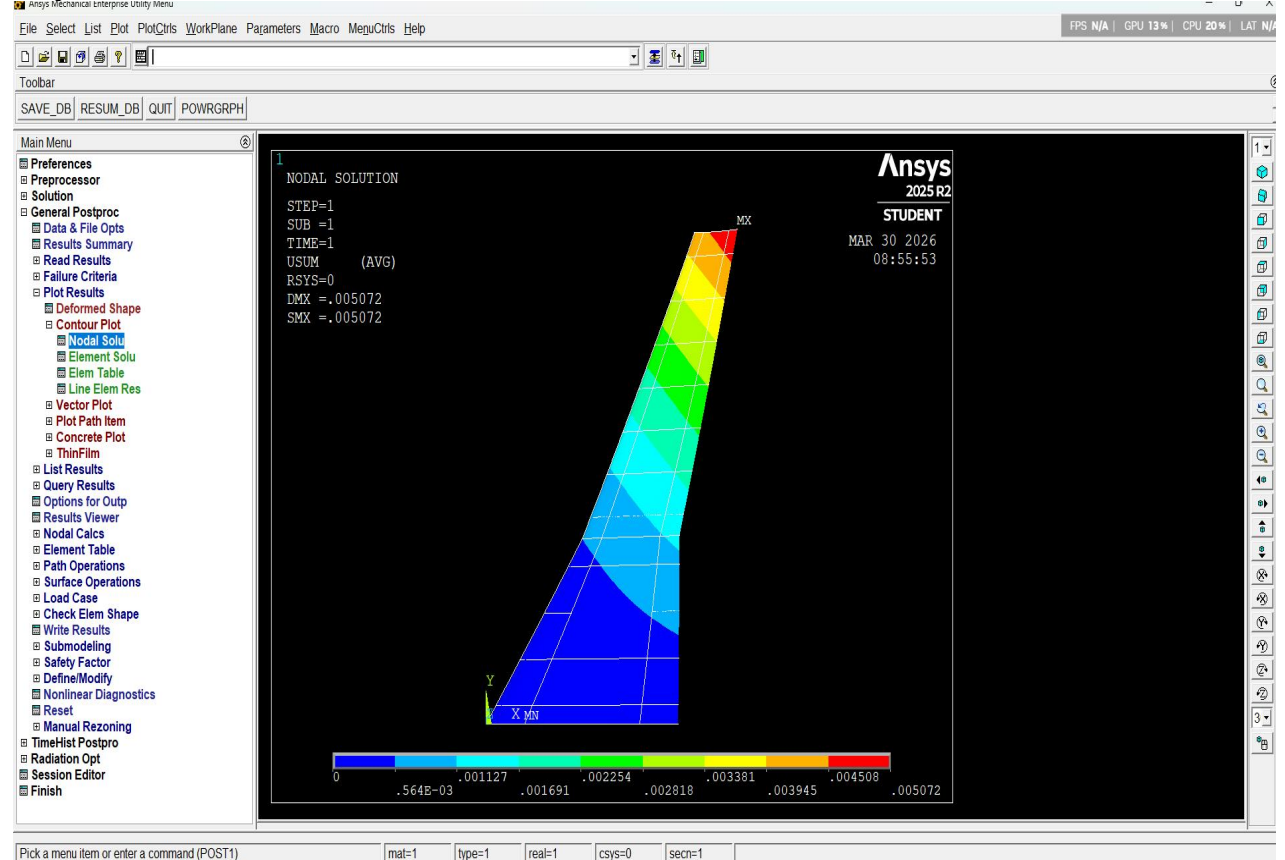
For Cantilever: Stress:



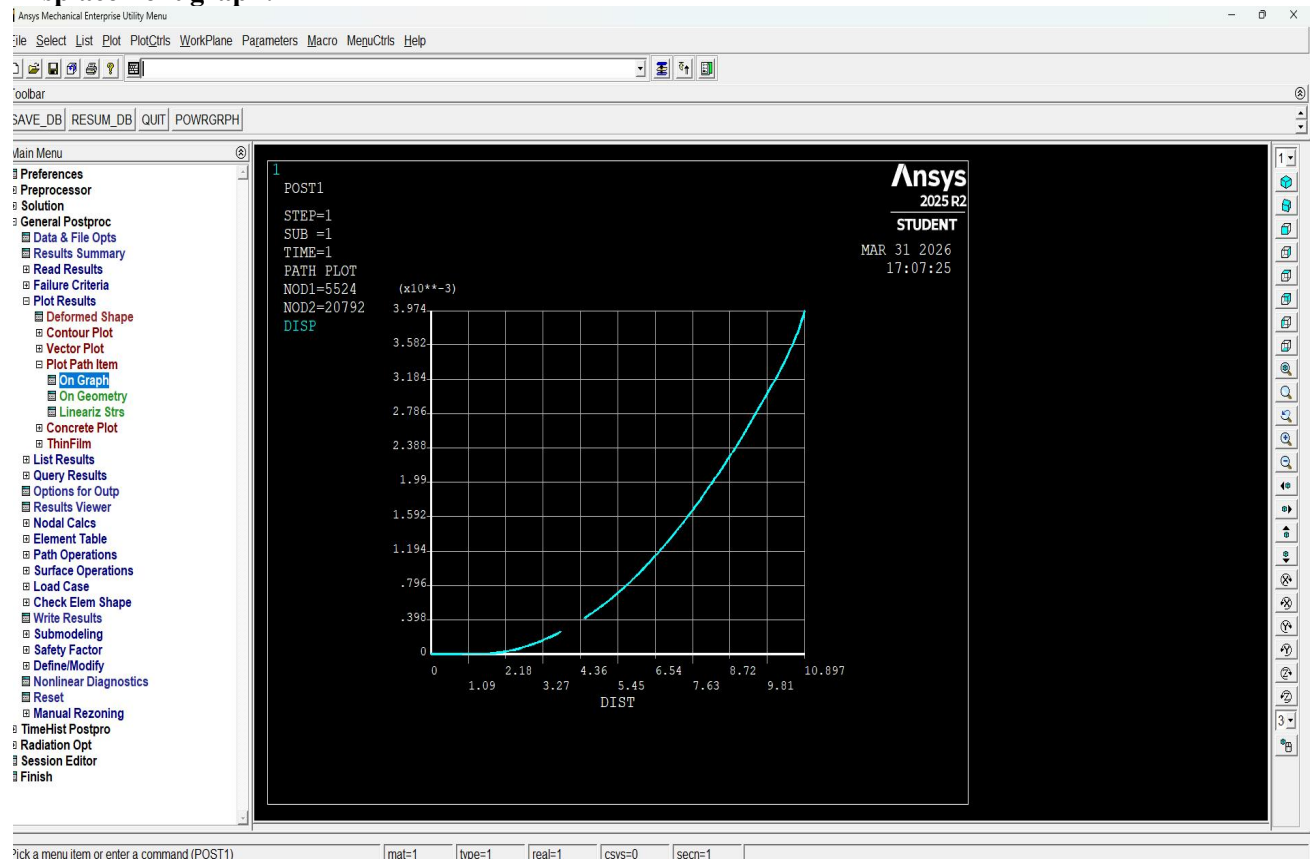
Stress graph:



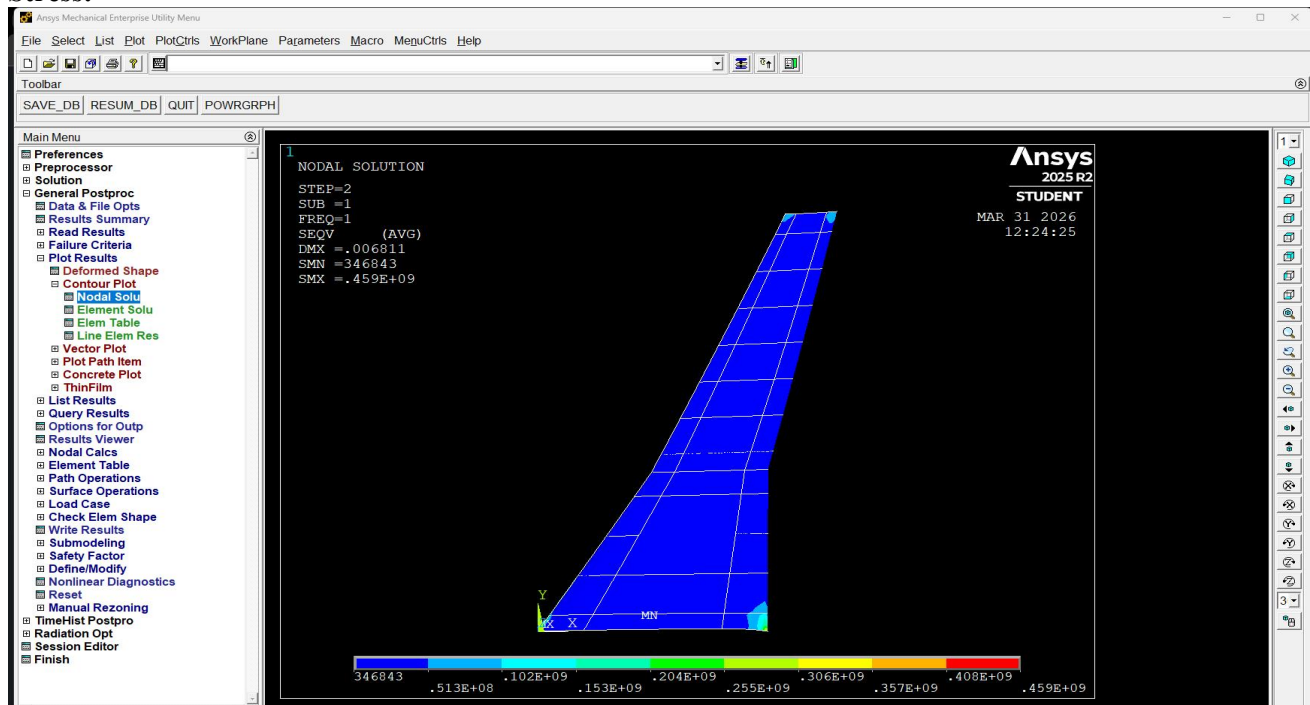
Displacement:



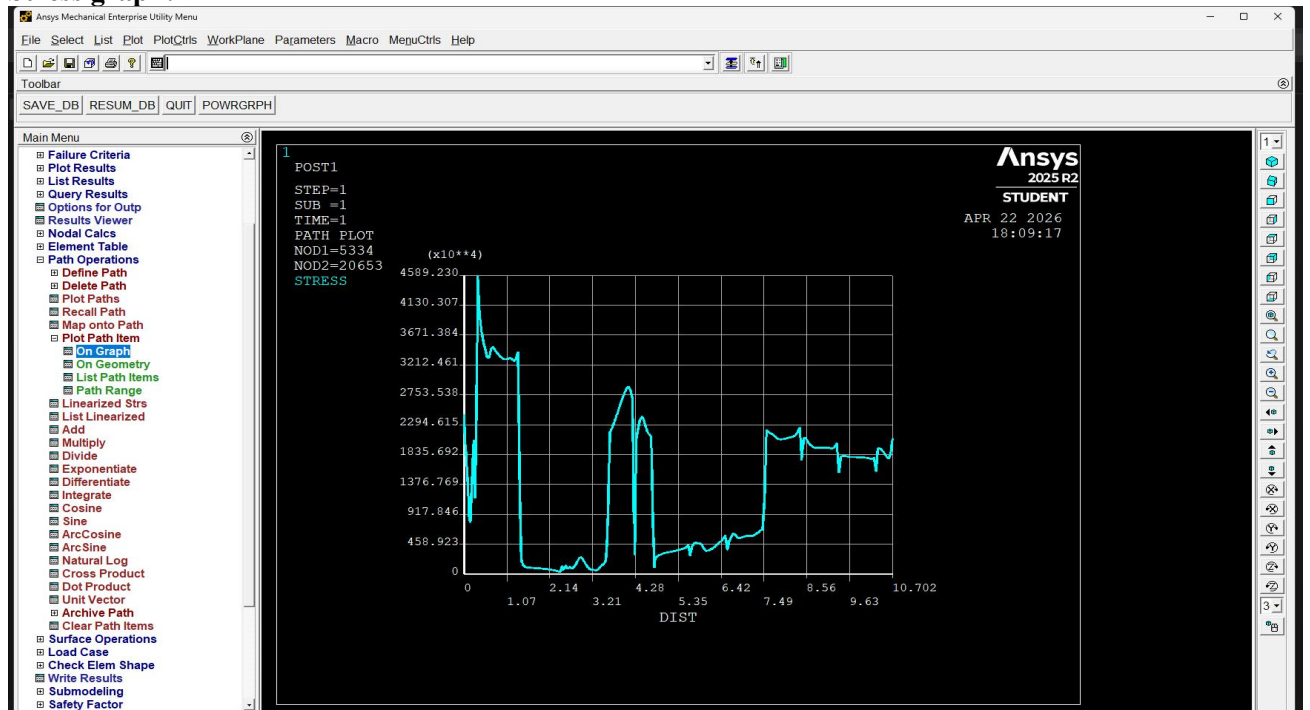
Displacement graph:



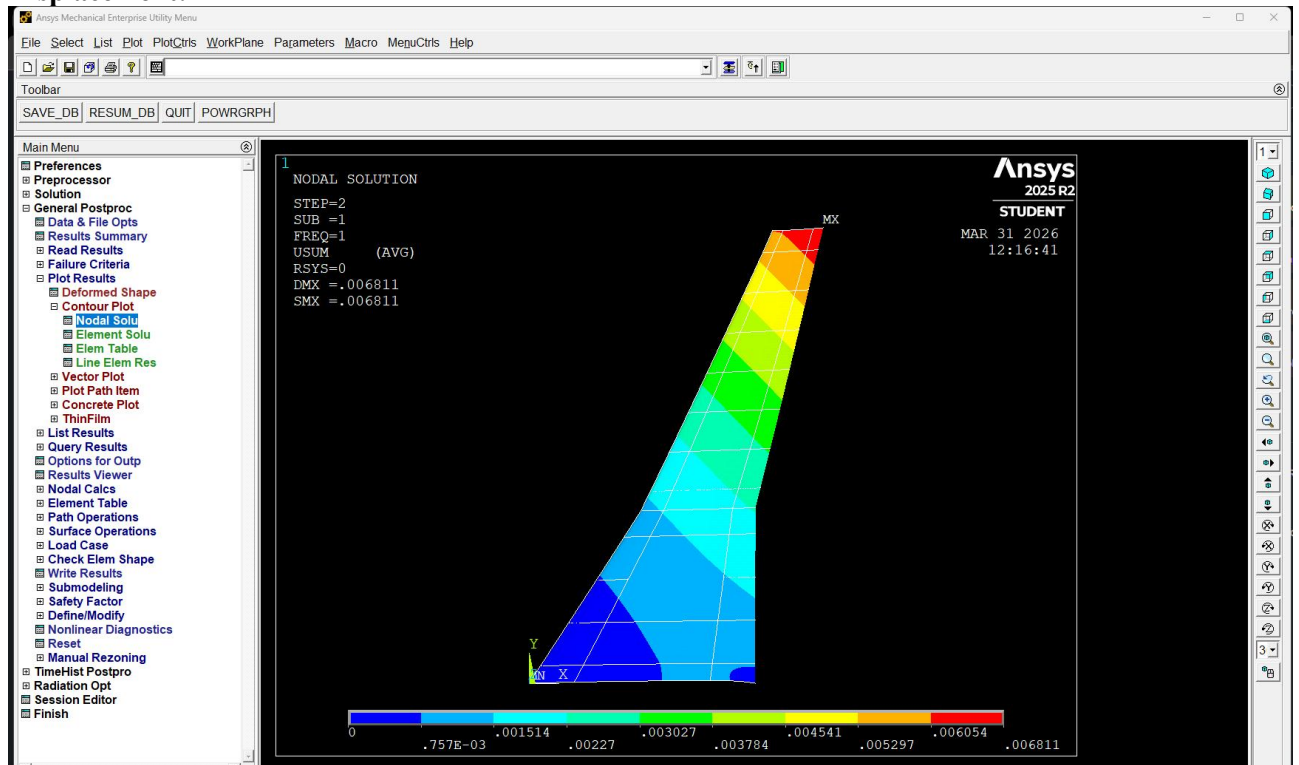
For Fixed ends: Stress:



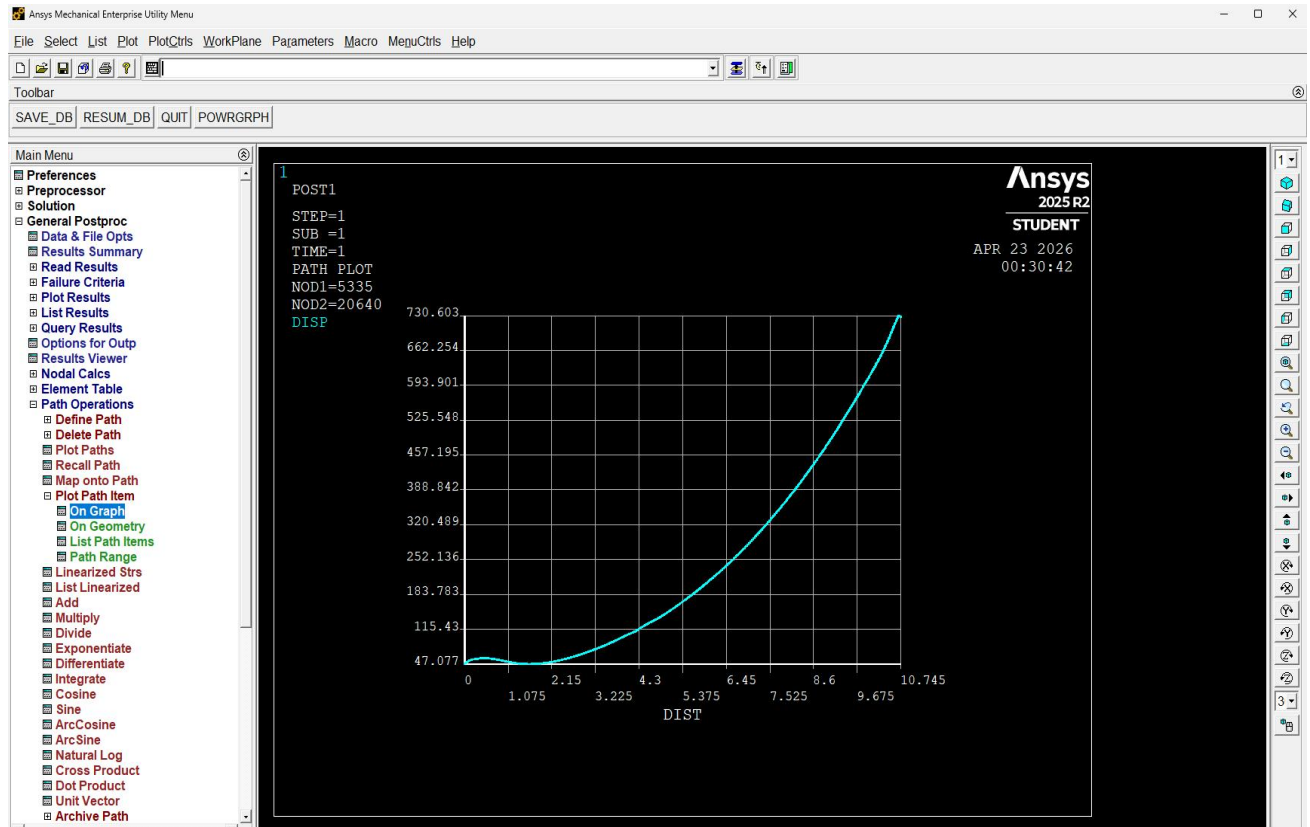
Stress graph:



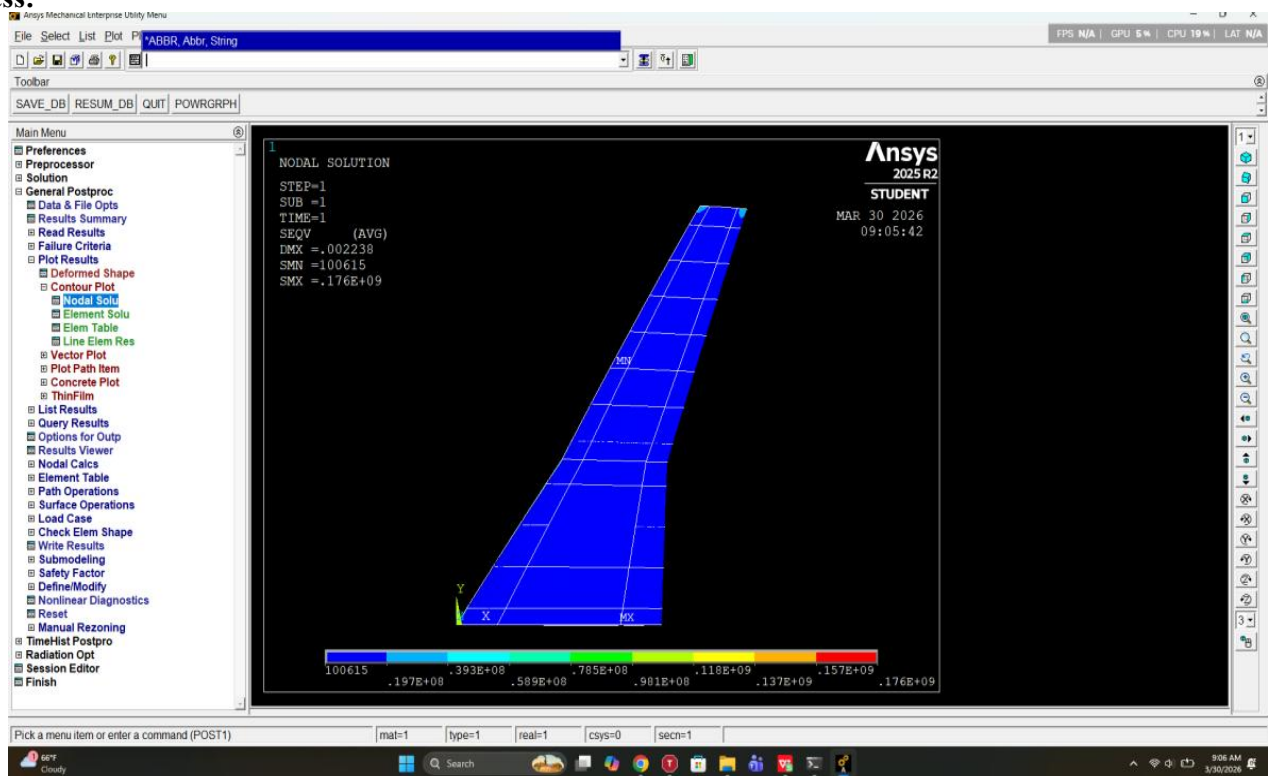
Displacement:



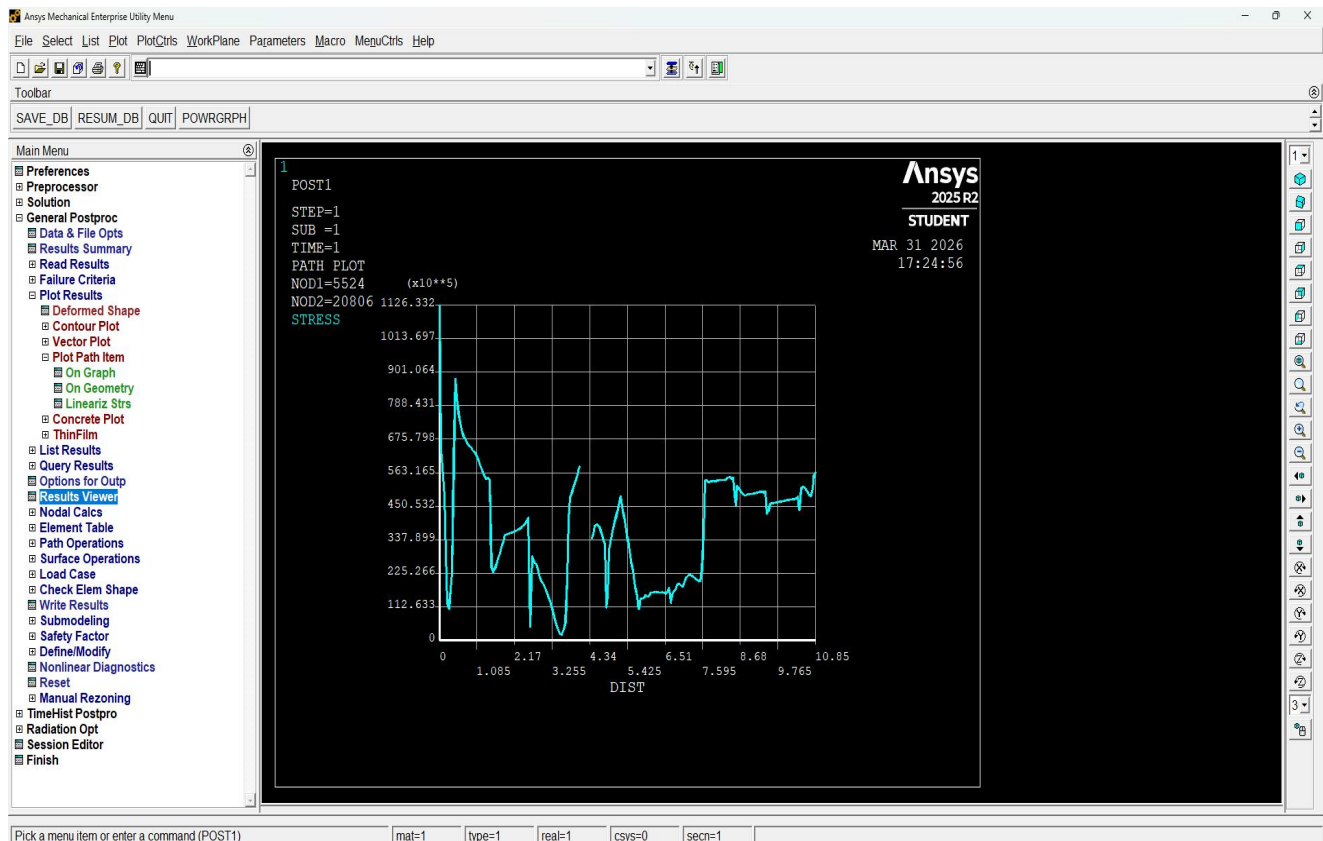
Displacement graph:



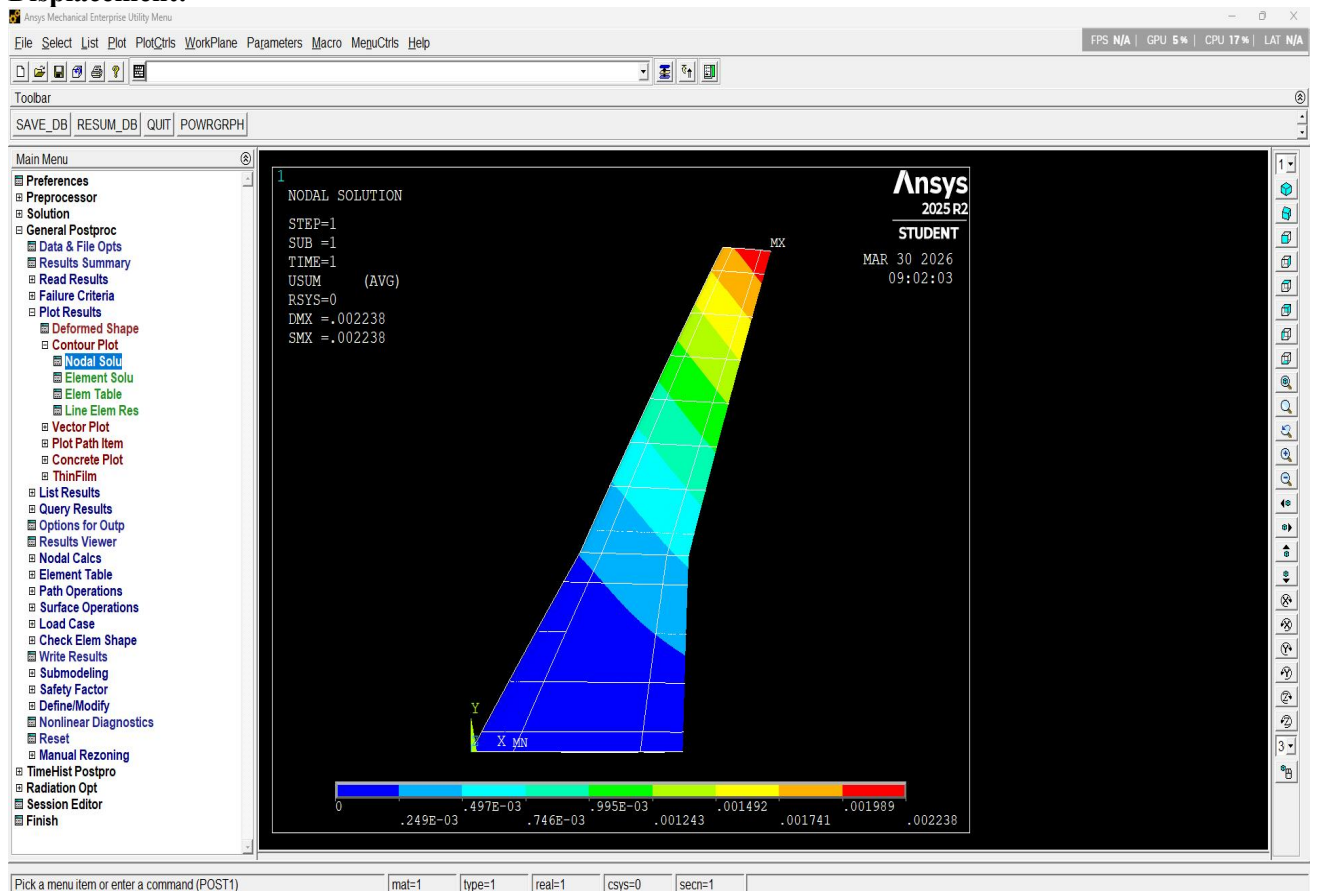
Wing-Down Load Case For Cantilever: Stress:



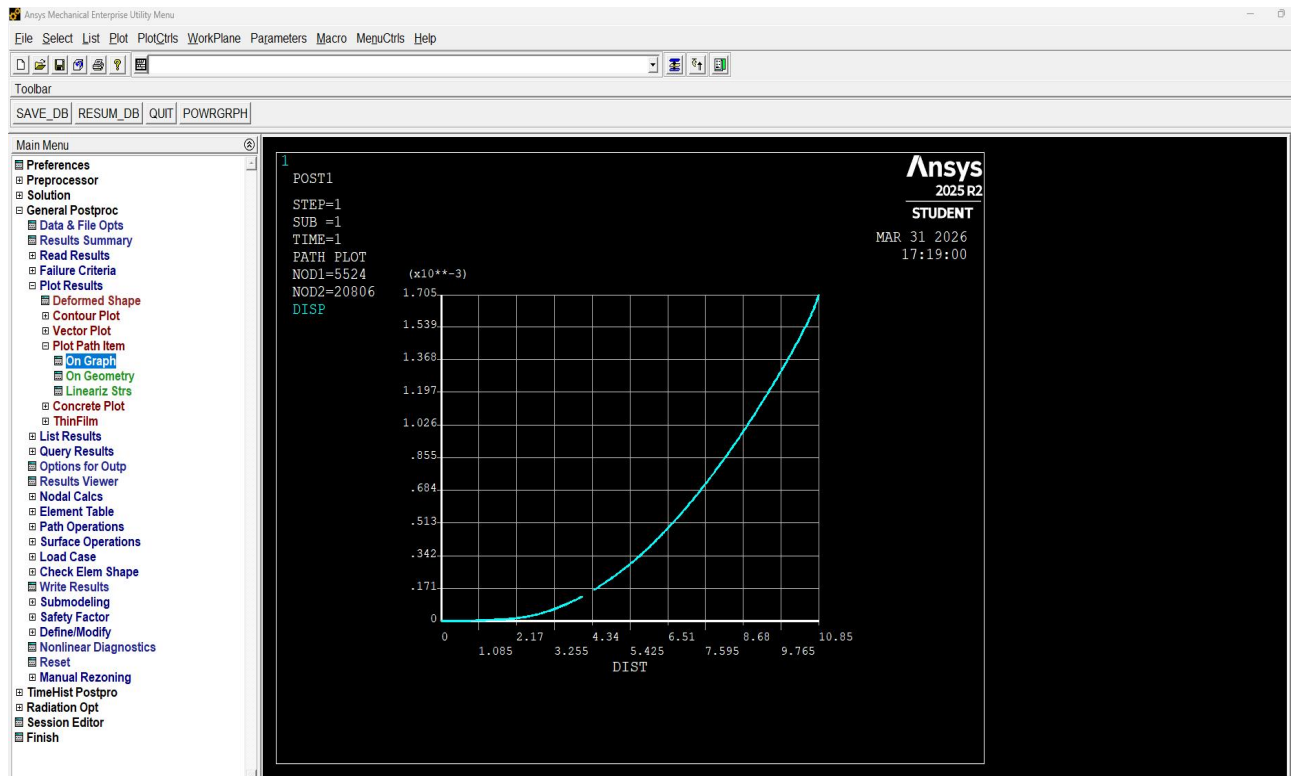
Stress graph:



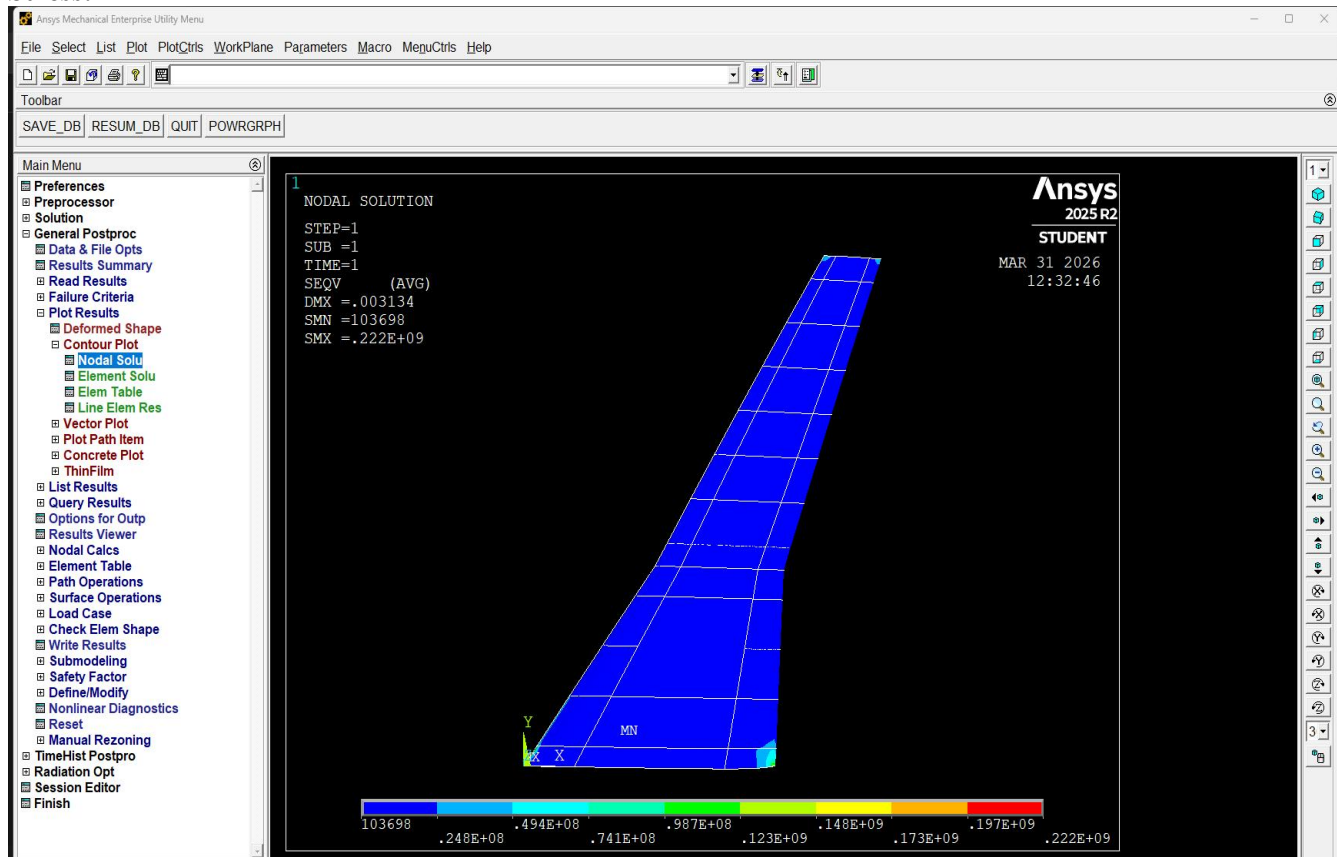
Displacement:



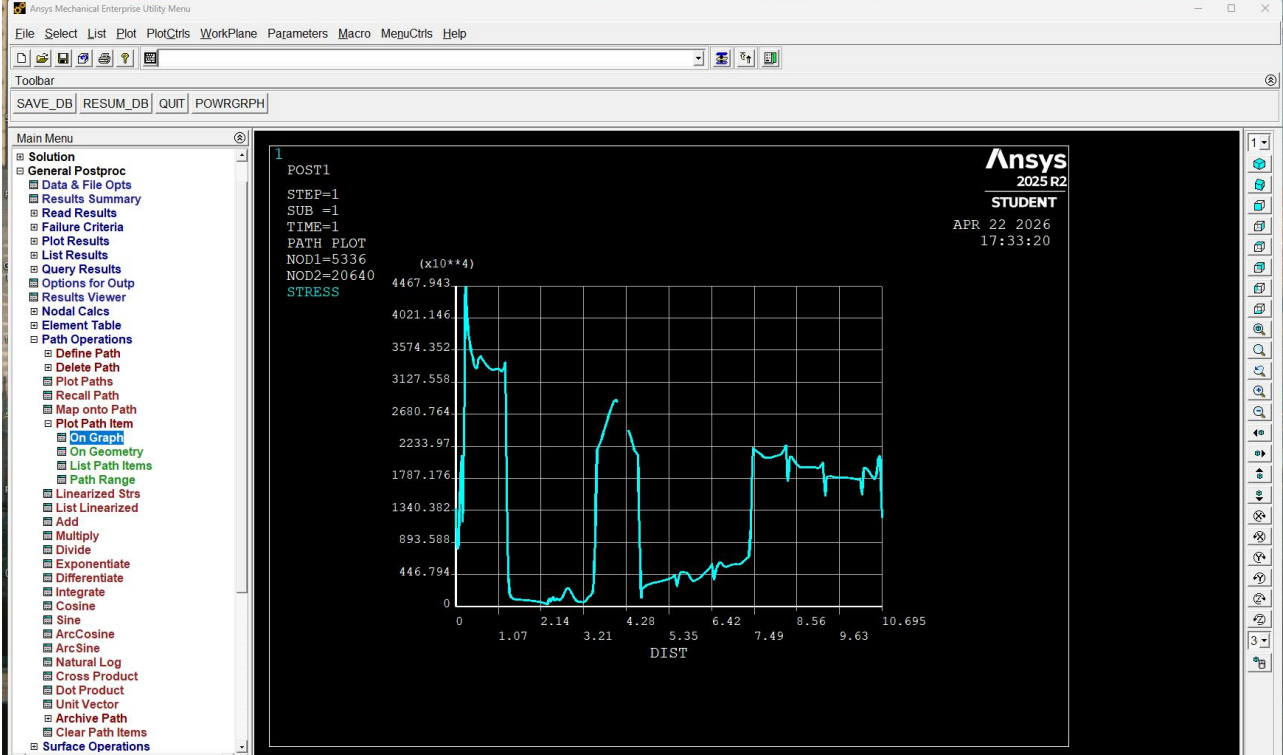
Displacement graph:



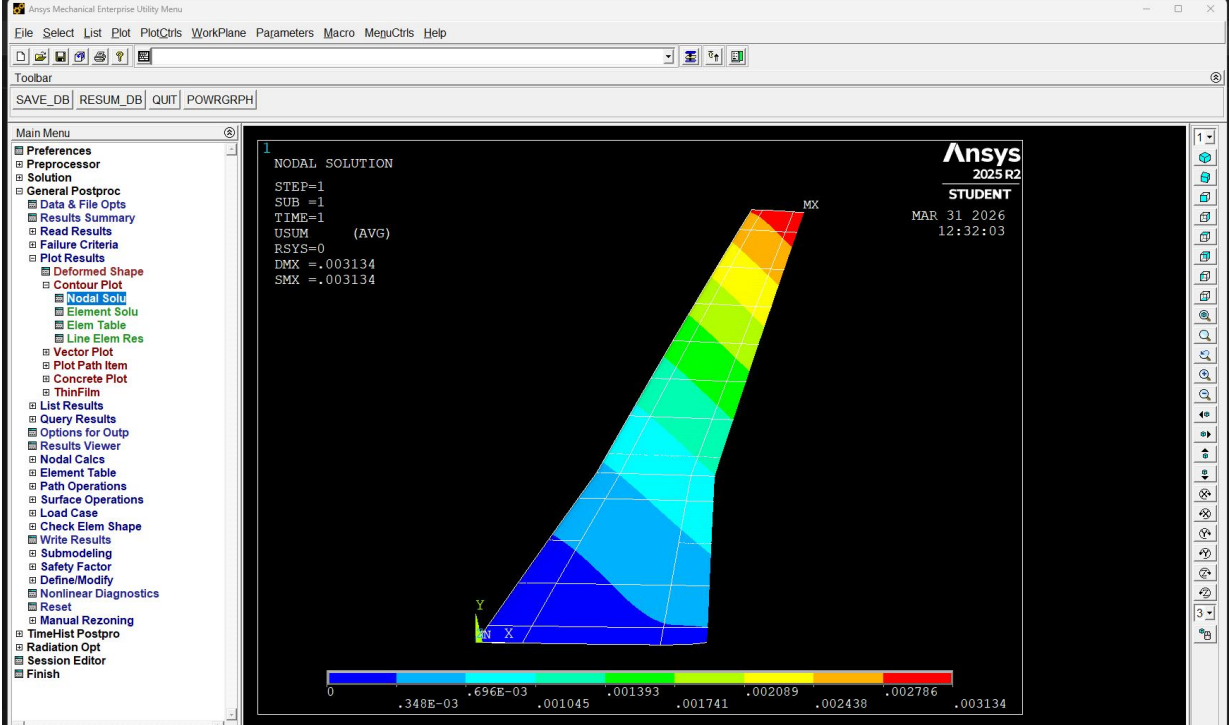
For Fixed ends: Stress:



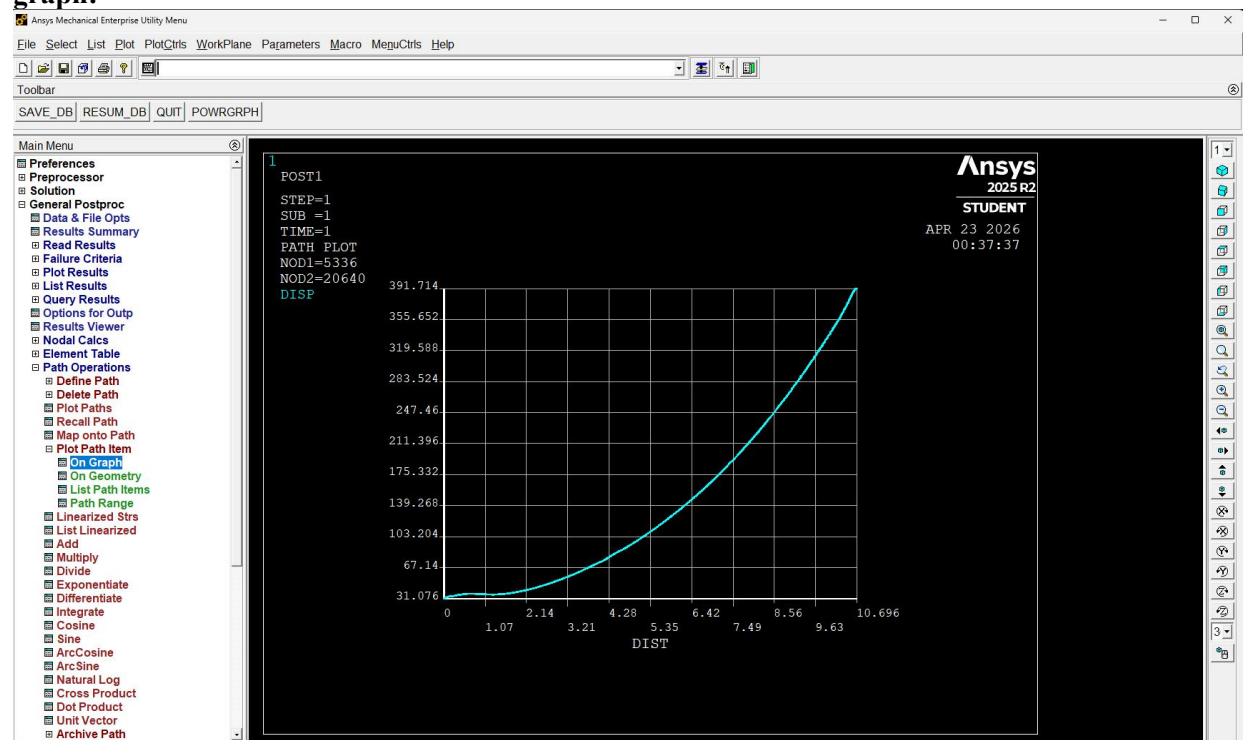
Stress graph:



Displacement:



Displacement graph:



1.3 Boundary Condition Comparison

To evaluate the effect of boundary constraints on the structural integrity of the wing box, the maximum stress and displacement values were compared between the standard Cantilever configuration (fixed only at the root, loads applied at the tip) and the Fixed Ends configuration (fixed at both the root and the tip, loads applied at mid-span).

The following values were recorded during the Wing-Down loading scenario:

Maximum von Mises Stress: Transitioning from a cantilever setup to a fixed-ends setup increased the maximum stress experienced by the wing box from 176 MPa ($.176\text{E}+09$ Pa) to 222 MPa ($.222\text{E}+09$ Pa)

Maximum Displacement: The structural deformation also increased under the fixed-ends constraint, with the maximum displacement shifting from 0.002238 m in the cantilever configuration to 0.003134 m in the fixed-ends configuration.

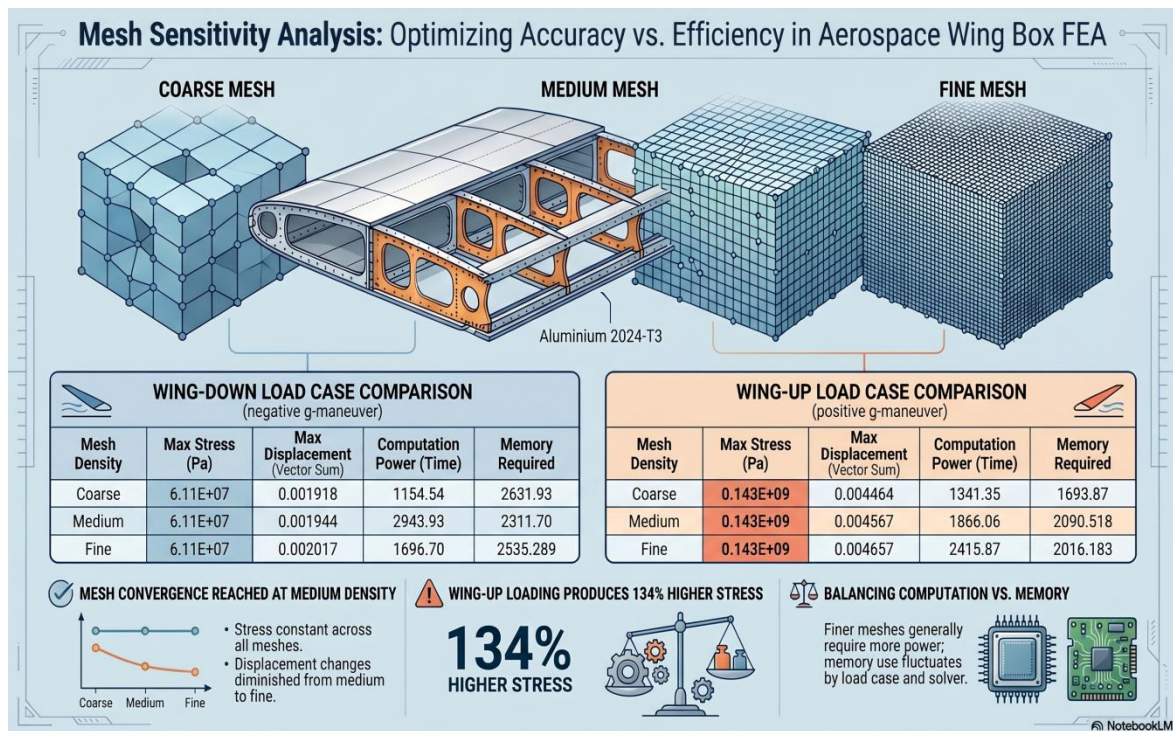
1.4 Yielding Confirmation

By utilizing the optimized structural thickness of 31.5 m, the structural integrity of the wing box was successfully maintained under all tested load cases. The absolute maximum von Mises stress recorded across all Section 1 tests occurred during the Wing-Up cantilever scenario, reaching $.340\text{E}+09$ Pa (340 MPa) and Because 340 MPa is safely below the established 345 MPa yield strength limit of the 2024-T3 Aluminium alloy, it is confirmed that the optimized design can withstand the extreme aerodynamic loads without experiencing any material yielding.

Section 2: Mesh Sensitivity Analysis

2.1 Mesh Parameter Data To investigate how mesh density affects the accuracy, convergence, and computational requirements of the finite element model, the wing box was analyzed using Coarse (0.1 m), Medium (0.05 m), and Fine (0.01 m) element sizes. The optimal 31.5 m structural thickness and linear Shell181 elements were maintained for all tests.

Table 2.1: Wing-Up and Wing-Down Load Case Results



2.2 Mesh Evaluation & Discussion;

Q: How does the mesh size affect the accuracy and convergence of the FEA results?

Ans. As the mesh transitioned from coarse to fine, the simulation demonstrated clear mesh convergence. The maximum von Mises stress remained highly stable across all mesh densities (.143E+09 Pa for Wing-Up and .611E+08 Pa for Wing-Down). However, the structural deformation results refined as the mesh size decreased. For instance, in the Wing-Up scenario, the maximum displacement changed from 0.004464 m (Coarse) to 0.004567 m (Medium), but the rate of change diminished when moving from Medium to Fine (0.004657 m). This diminishing change confirms the results are converging on an accurate solution.

Q: How does the mesh size affect the computational time and memory required for the FEA analysis?

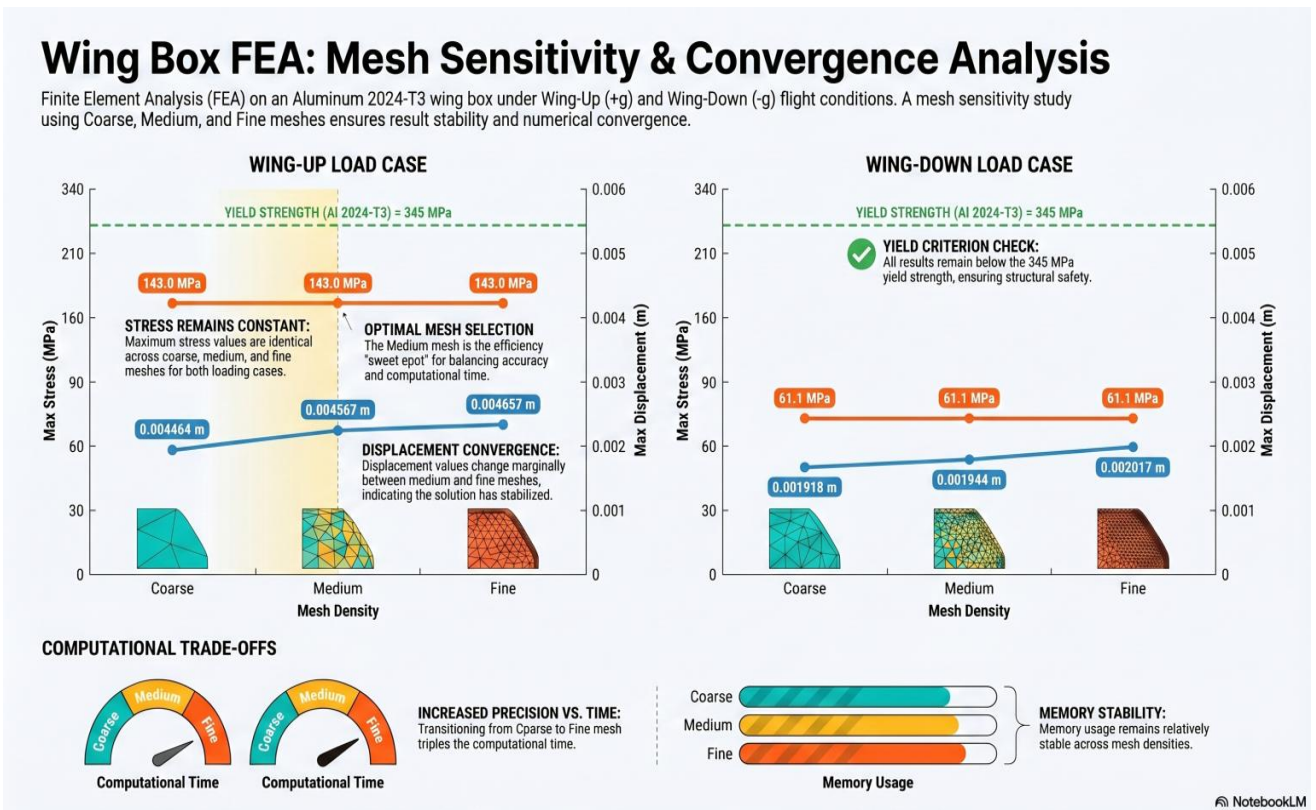
Ans. Finer meshes exponentially increase the number of nodes and elements, which significantly raises computational demands. In the Wing-Up case, computation time nearly doubled from 1341.35 seconds for the coarse mesh to 2415.87 seconds for the fine mesh. Memory requirements also increased from 1693.87 MB to over 2016.18 MB.

Q: What is the optimal mesh size that balances accuracy and efficiency for your FEA model?

Ans. The optimal mesh size for this model is the Medium mesh (0.05 m). It produced highly accurate, converged displacement results at a reasonable computational cost, avoiding the excessive processing time required by the fine mesh. The medium mesh will be utilized for the remainder of the analysis.

2.3 Visual Results

Mesh Comparison Data:

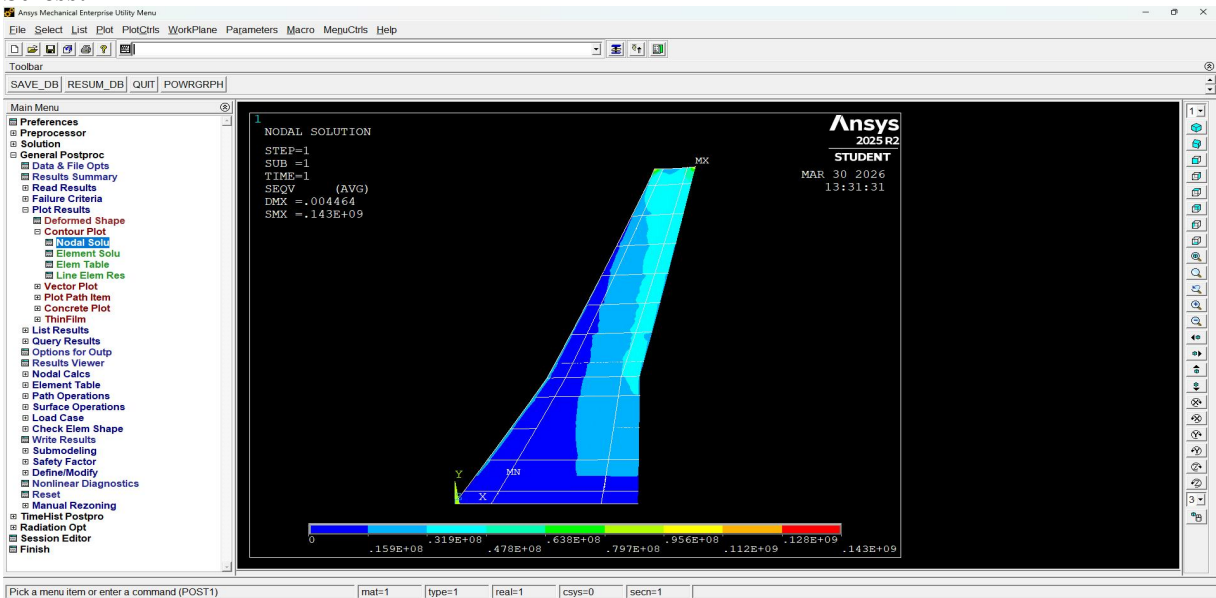


Since medium mesh was used for Section 1 analysis and contours were attached above. Only coarse and fine mesh contours will be displayed.

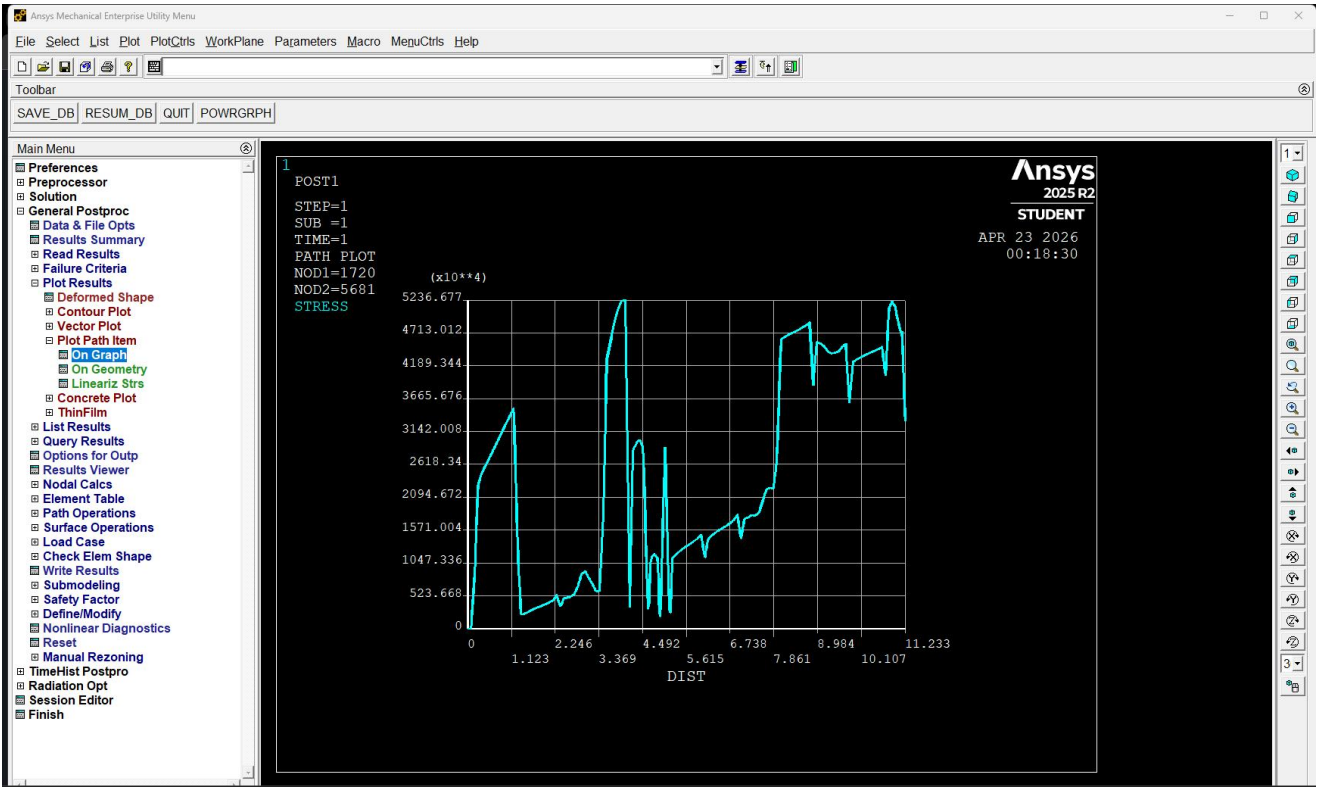
Wing-Up Load Case Contours:

Coarse Mesh:

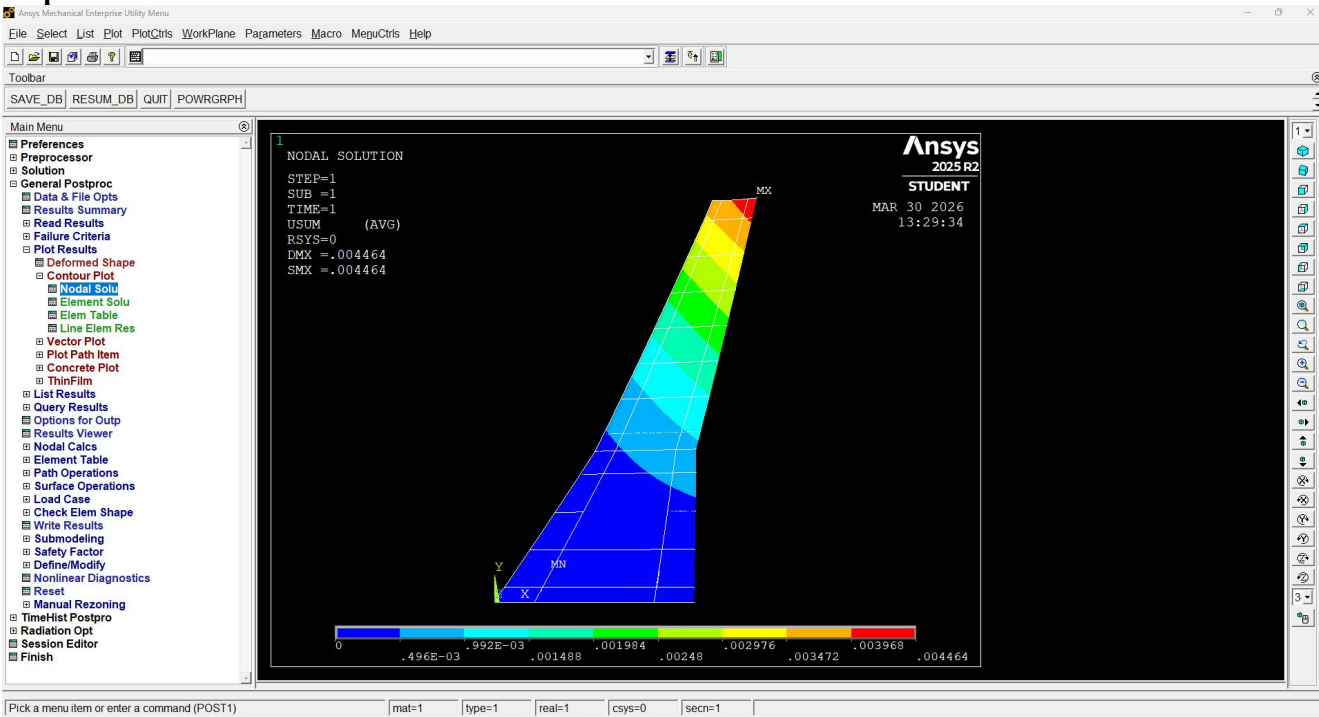
Stress:



Stress graph:



Displacement:



Computational Power and Memory:

```
DISTRIBUTED SPARSE MATRIX DIRECT SOLVER.
Number of equations = 125778, Maximum wavefront = 96

Memory allocated on only this MPI rank (rank 0)
-----
Equation solver memory allocated = 122.624 MB
Equation solver memory required for in-core mode = 117.791 MB
Equation solver memory required for out-of-core mode = 54.780 MB
Total (solver and non-solver) memory allocated = 708.284 MB

Total memory summed across all MPI ranks on this machines
-----
Equation solver memory allocated = 482.251 MB
Equation solver memory required for in-core mode = 462.562 MB
Equation solver memory required for out-of-core mode = 194.884 MB
Total (solver and non-solver) memory allocated = 1693.079 MB

*** NOTE *** CP = 1338.609 TIME= 10:09:58
The Distributed Sparse Matrix Solver is currently running in the
in-core memory mode. This memory mode uses the most amount of memory
in order to avoid using the hard drive as much as possible, which most
often results in the fastest solution time. This mode is recommended
if enough physical memory is present to accommodate all of the solver
data.
curEqn= 30786 totEqn= 30786 Job CP sec= 1340.016
Factor Done= 100% Factor Wall sec= 1.042 rate= 4.8 GFlops
Distributed sparse solver maximum pivot= 4.954156527E+16 at node 1947
UX.
Distributed sparse solver minimum pivot= 4.713410629E+11 at node 16337
ROTZ.
Distributed sparse solver minimum pivot in absolute value=
4.713410629E+11 at node 16337 ROTZ.

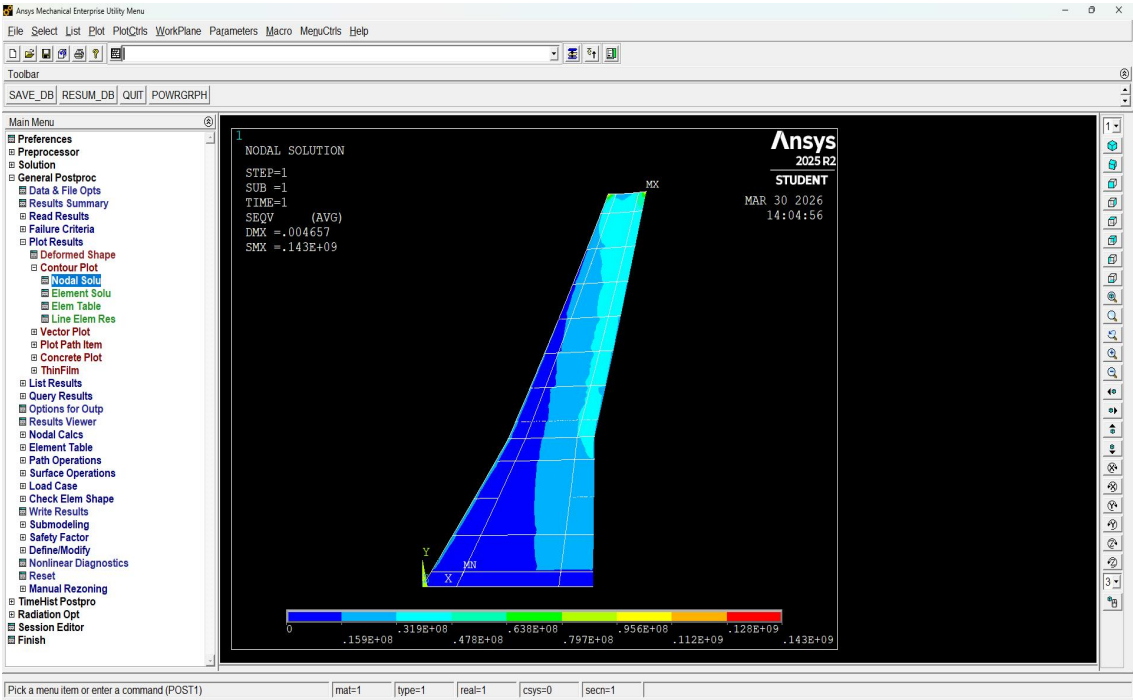
*** ELEMENT RESULT CALCULATION TIMES
TYPE NUMBER ENAME TOTAL CP AVE CP
1 25316 SHELL181 2.859 0.000113

*** NODAL LOAD CALCULATION TIMES
TYPE NUMBER ENAME TOTAL CP AVE CP
1 25316 SHELL181 0.359 0.000014
*** LOAD STEP 1 SUBSTEP 1 COMPLETED. CUM ITER = 1
*** TIME = 1.00000 TIME INC = 1.00000 NEW TRIANG MATRIX

*** NOTE *** CP = 1341.359 TIME= 10:10:01
Solution is done!
```

Fine mesh:

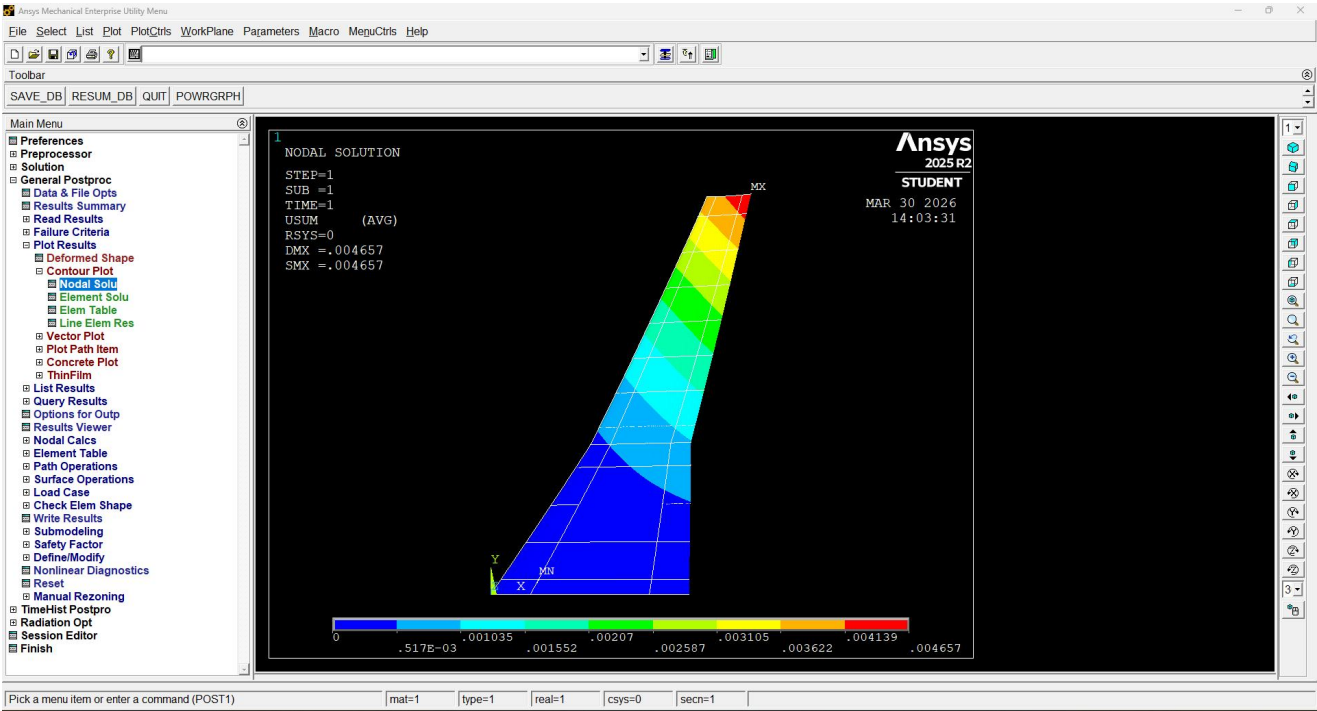
Stress:



Stress graph:



Displacement:



Computational Power and Memory:

```
Total memory summed across all MPI ranks on this machines
-----
Equation solver memory allocated           = 434.695 MB
Equation solver memory required for in-core mode = 417.006 MB
Equation solver memory required for out-of-core mode = 178.767 MB
Total (solver and non-solver) memory allocated = 2016.183 MB

*** NOTE ***                      CP = 2413.781    TIME= 11:15:30
The Distributed Sparse Matrix Solver is currently running in the
in-core memory mode. This memory mode uses the most amount of memory
in order to avoid using the hard drive as much as possible, which most
often results in the fastest solution time. This mode is recommended
if enough physical memory is present to accommodate all of the solver
data.
curEqn= 29304 totEqn= 29304 Job CP sec= 2414.766
Factor Done= 100% Factor Wall sec= 0.788 rate= 6.2 GFlops
Distributed sparse solver maximum pivot= 4.797458244E+16 at node 1950
UX.
Distributed sparse solver minimum pivot= 3.958363127E+11 at node 24140
ROTZ.
Distributed sparse solver minimum pivot in absolute value=
3.958363127E+11 at node 24140 ROTZ.

*** ELEMENT RESULT CALCULATION TIMES
TYPE    NUMBER    ENAME    TOTAL CP    AVE CP
1       25316    SHELL181    2.250    0.000089

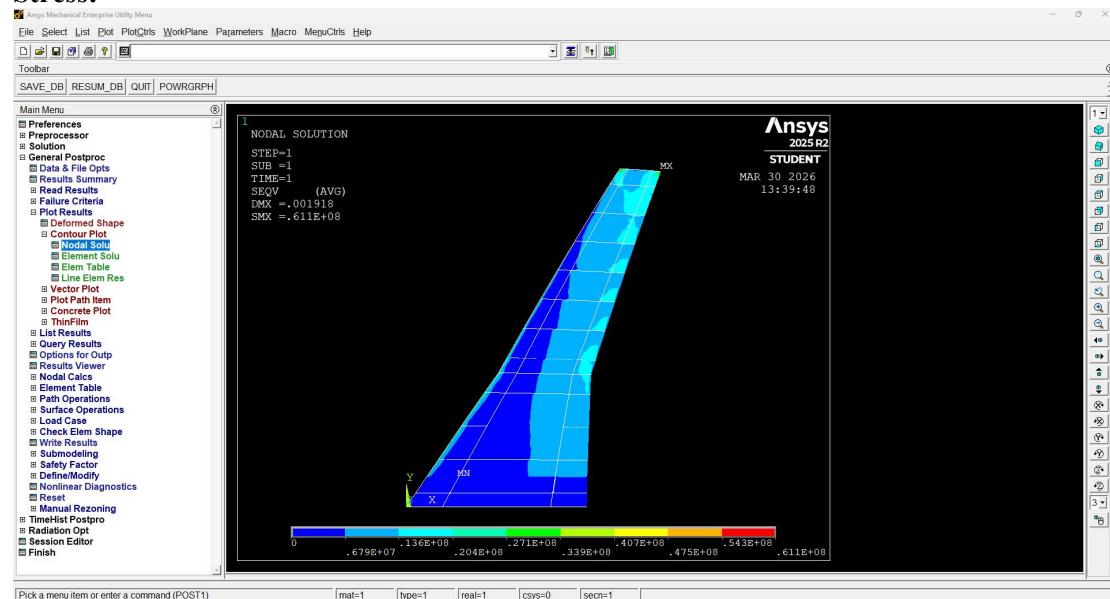
*** NODAL LOAD CALCULATION TIMES
TYPE    NUMBER    ENAME    TOTAL CP    AVE CP
1       25316    SHELL181    0.312    0.000012
*** LOAD STEP    1    SUBSTEP    1    COMPLETED.    CUM ITER =    1
*** TIME = 1.00000    TIME INC = 1.00000    NEW TRIANG MATRIX

*** NOTE ***                      CP = 2415.875    TIME= 11:15:32
Solution is done!
```

Wing-Down Load Case Contours:

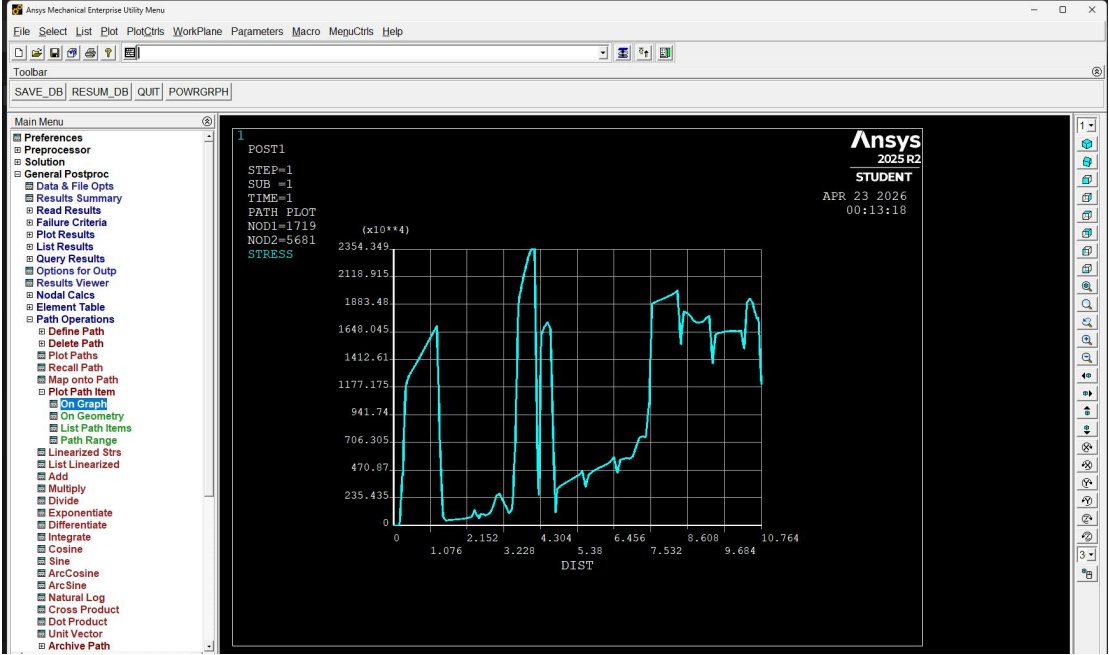
Coarse mesh:

Stress:

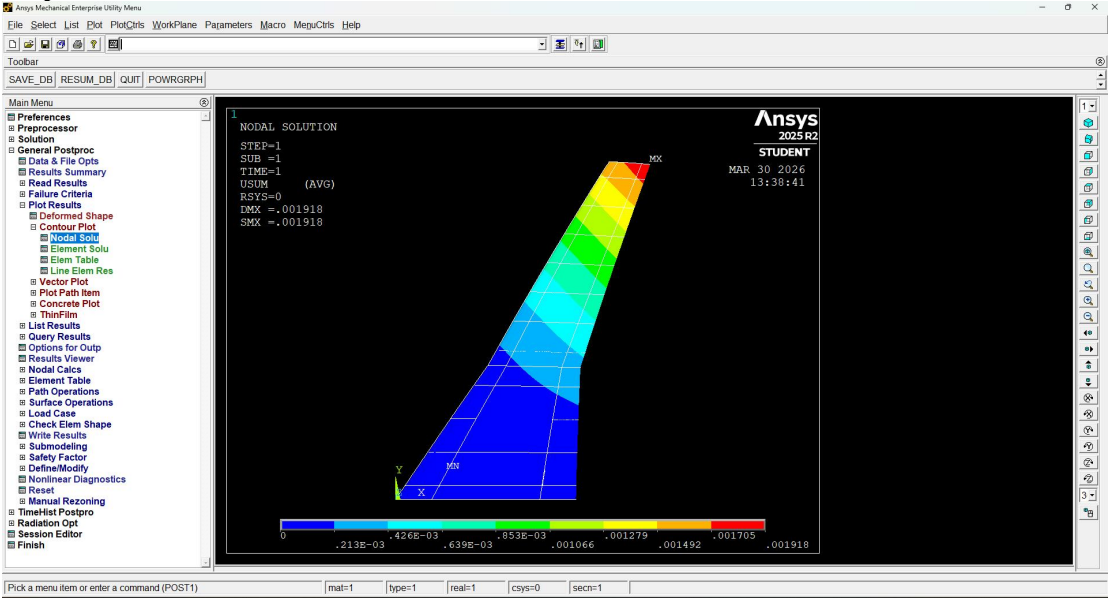


Stress

graph:



Displacement:



Computational Power and Memory:

```
Memory allocated on only this MPI rank (rank 0)
-----
Equation solver memory allocated = 110.017 MB
Equation solver memory required for in-core mode = 105.742 MB
Equation solver memory required for out-of-core mode = 50.625 MB
Total (solver and non-solver) memory allocated = 1147.998 MB

Total memory summed across all MPI ranks on this machines
-----
Equation solver memory allocated = 473.205 MB
Equation solver memory required for in-core mode = 453.844 MB
Equation solver memory required for out-of-core mode = 188.593 MB
Total (solver and non-solver) memory allocated = 2361.939 MB

*** NOTE ***
CP = 1152.281 TIME= 13:38:23
The Distributed Sparse Matrix Solver is currently running in the
in-core memory mode. This memory mode uses the most amount of memory
in order to avoid using the hard drive as much as possible, which most
often results in the fastest solution time. This mode is recommended
if enough physical memory is present to accommodate all of the solver
data.
curEqn= 26724 totEqn= 26724 Job CP sec= 1153.250
Factor Done= 100% Factor Wall sec= 0.724 rate= 5.9 GFlops
Distributed sparse solver maximum pivot= 5.053978713E+16 at node 1934
UX.
Distributed sparse solver minimum pivot= 4.284765216E+11 at node 19128
ROTZ.
Distributed sparse solver minimum pivot in absolute value=
4.284765216E+11 at node 19128 ROTZ.

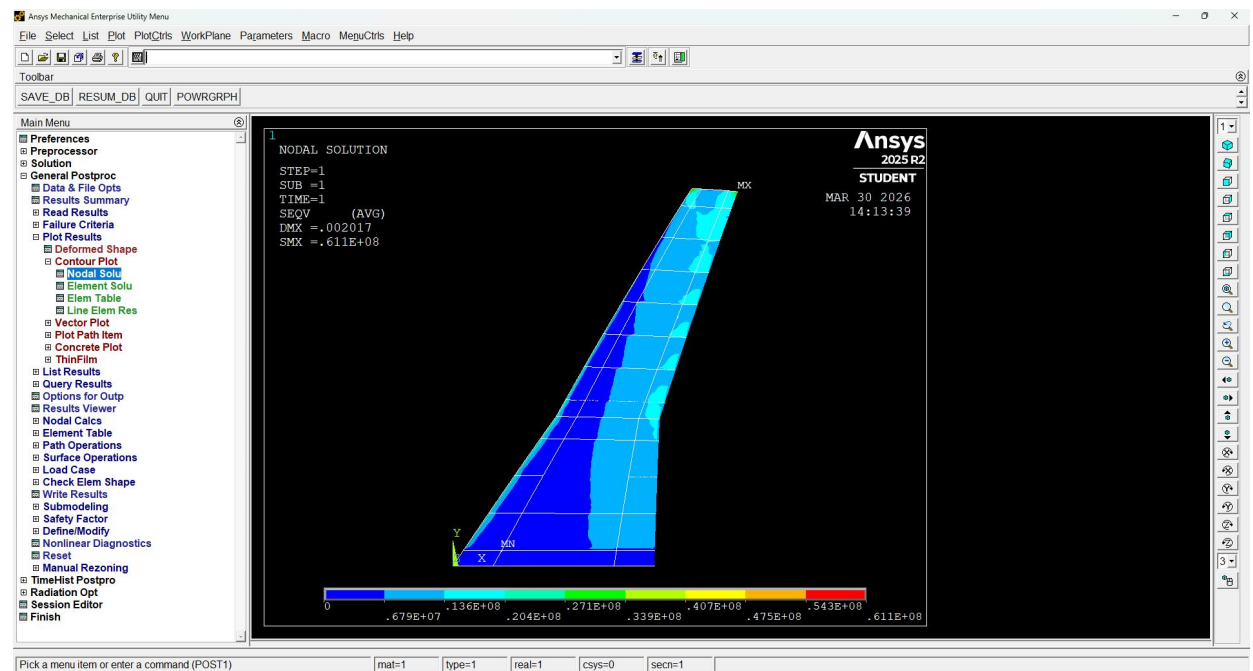
*** ELEMENT RESULT CALCULATION TIMES
TYPE NUMBER ENAME TOTAL CP AVE CP
1 25316 SHELL181 2.750 0.000109

*** NODAL LOAD CALCULATION TIMES
TYPE NUMBER ENAME TOTAL CP AVE CP
1 25316 SHELL181 0.328 0.000013
*** LOAD STEP 1 SUBSTEP 1 COMPLETED. CUM ITER = 1
*** TIME = 1.00000 TIME INC = 1.00000 NEW TRIANG MATRIX

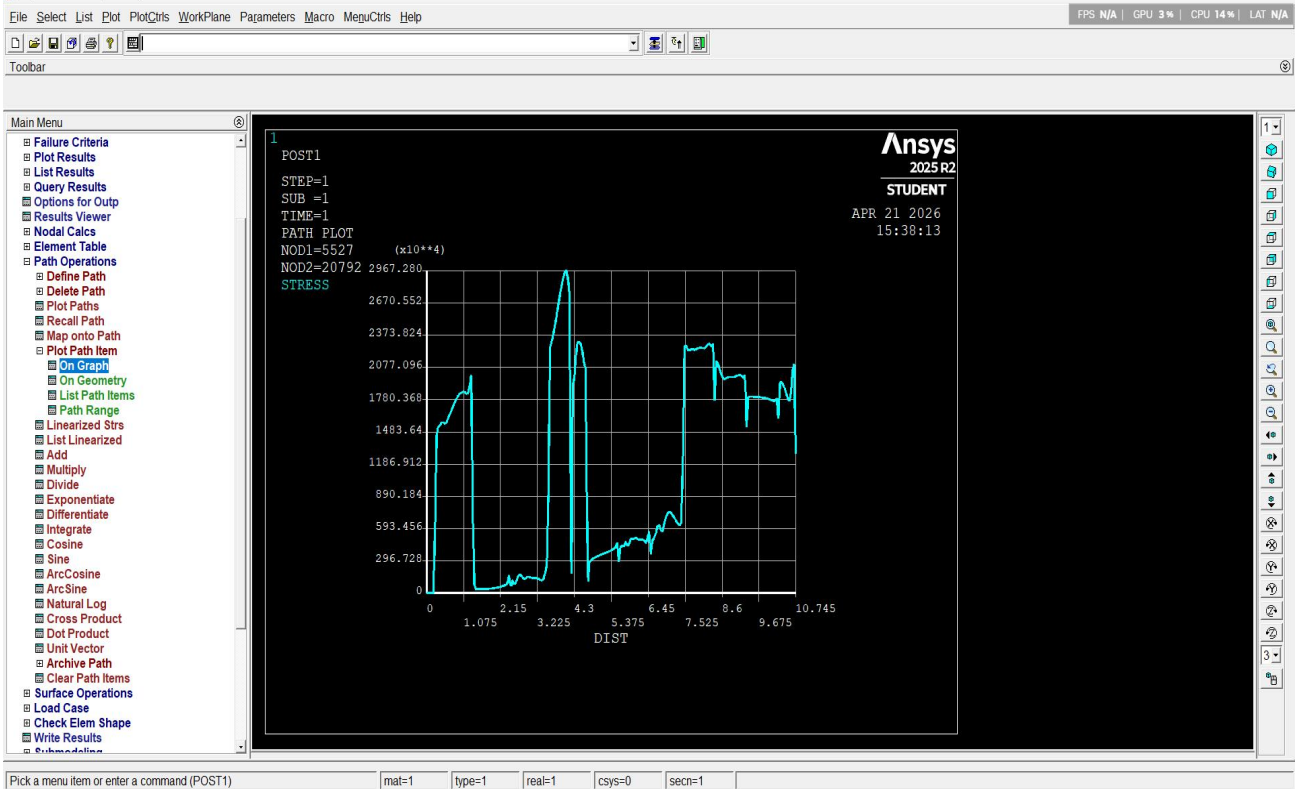
*** NOTE ***
CP = 1154.547 TIME= 13:38:25
Solution is done!
```

Fine mesh:

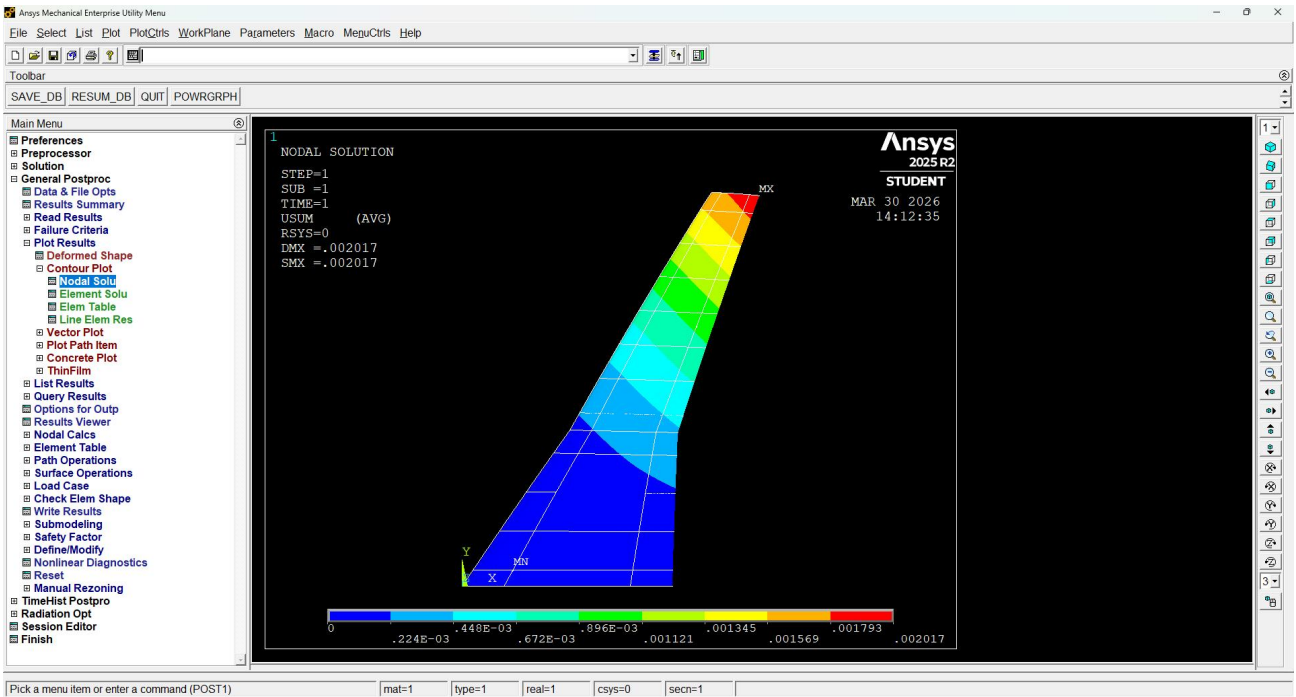
Stress:



Stress
graph:



Displacement:



Computational Power and Memory:

```
Total memory summed across all MPI ranks on this machines
-----
Equation solver memory allocated          = 504.257 MB
Equation solver memory required for in-core mode = 483.658 MB
Equation solver memory required for out-of-core mode = 203.216 MB
Total (solver and non-solver) memory allocated = 2535.289 MB

*** NOTE ***                      CP = 1694.328    TIME= 14:11:10
The Distributed Sparse Matrix Solver is currently running in the
in-core memory mode. This memory mode uses the most amount of memory
in order to avoid using the hard drive as much as possible, which most
often results in the fastest solution time. This mode is recommended
if enough physical memory is present to accommodate all of the solver
data.
curEqn= 31476 totEqn= 31476 Job CP sec= 1695.422
Factor Done= 100% Factor Wall sec= 0.870 rate= 6.1 GFlops
Distributed sparse solver maximum pivot= 4.951863164E+16 at node 1947
UX.
Distributed sparse solver minimum pivot= 4.644583988E+11 at node 21457
ROTZ.
Distributed sparse solver minimum pivot in absolute value=
4.644583988E+11 at node 21457 ROTZ.

*** ELEMENT RESULT CALCULATION TIMES
TYPE      NUMBER      ENAME      TOTAL CP      AVE CP
1         25316      SHELL181      2.797      0.000110

*** NODAL LOAD CALCULATION TIMES
TYPE      NUMBER      ENAME      TOTAL CP      AVE CP
1         25316      SHELL181      0.594      0.000023
*** LOAD STEP 1 SUBSTEP 1 COMPLETED. CUM ITER = 1
*** TIME = 1.00000 TIME INC = 1.00000 NEW TRIANG MATRIX

*** NOTE ***                      CP = 1696.703    TIME= 14:11:13
Solution is done!
```

For Medium mesh Displacement and Stress contours have been documented in Section 1.

Hence only computational power remains which is attached below.

For Wing-up:

```
Memory allocated on only this MPI rank (rank 0)
-----
Equation solver memory allocated          = 161.005 MB
Equation solver memory required for in-core mode = 154.442 MB
Equation solver memory required for out-of-core mode = 63.380 MB
Total (solver and non-solver) memory allocated = 978.056 MB

Total memory summed across all MPI ranks on this machines
-----
Equation solver memory allocated          = 572.026 MB
Equation solver memory required for in-core mode = 548.568 MB
Equation solver memory required for out-of-core mode = 224.825 MB
Total (solver and non-solver) memory allocated = 2090.518 MB

*** NOTE ***                      CP = 1863.281    TIME= 10:46:41
The Distributed Sparse Matrix Solver is currently running in the
in-core memory mode. This memory mode uses the most amount of memory
in order to avoid using the hard drive as much as possible, which most
often results in the fastest solution time. This mode is recommended
if enough physical memory is present to accommodate all of the solver
data.
curEqn= 39162 totEqn= 39162 Job CP sec= 1864.750
Factor Done= 100% Factor Wall sec= 1.205 rate= 6.4 GFlops
Distributed sparse solver maximum pivot= 5.104195334E+16 at node 1930
UX.
Distributed sparse solver minimum pivot= 1.786201763E+11 at node 5135
ROTZ.
Distributed sparse solver minimum pivot in absolute value=
1.786201763E+11 at node 5135 ROTZ.

*** ELEMENT RESULT CALCULATION TIMES
TYPE      NUMBER      ENAME      TOTAL CP      AVE CP
1         25316      SHELL181      2.734      0.000108

*** NODAL LOAD CALCULATION TIMES
TYPE      NUMBER      ENAME      TOTAL CP      AVE CP
1         25316      SHELL181      0.406      0.000016
*** LOAD STEP 1 SUBSTEP 1 COMPLETED. CUM ITER = 1
*** TIME = 1.00000 TIME INC = 1.00000 NEW TRIANG MATRIX

*** NOTE ***                      CP = 1866.062    TIME= 10:46:44
Solution is done!
```

For Wing-down:

```
Memory allocated on only this MPI rank (rank      0)
-----
Equation solver memory allocated                = 161.005 MB
Equation solver memory required for in-core mode = 154.442 MB
Equation solver memory required for out-of-core mode = 63.380 MB
Total (solver and non-solver) memory allocated = 1063.267 MB

Total memory summed across all MPI ranks on this machines
-----
Equation solver memory allocated                = 572.026 MB
Equation solver memory required for in-core mode = 548.568 MB
Equation solver memory required for out-of-core mode = 224.825 MB
Total (solver and non-solver) memory allocated = 2311.729 MB

*** NOTE ***
CP = 2940.984 TIME= 11:44:47
The Distributed Sparse Matrix Solver is currently running in the
in-core memory mode. This memory mode uses the most amount of memory
in order to avoid using the hard drive as much as possible, which most
often results in the fastest solution time. This mode is recommended
if enough physical memory is present to accommodate all of the solver
data.
curEqn= 39162 totEqn= 39162 Job CP sec= 2942.484
Factor Done= 100% Factor Wall sec= 1.201 rate= 6.4 GFlops
Distributed sparse solver maximum pivot= 5.104195334E+16 at node 1930
UX.
Distributed sparse solver minimum pivot= 1.786201763E+11 at node 5135
ROTZ.
Distributed sparse solver minimum pivot in absolute value=
1.786201763E+11 at node 5135 ROTZ.

*** ELEMENT RESULT CALCULATION TIMES
TYPE NUMBER ENAME TOTAL CP AVE CP
1 25316 SHELL181 2.844 0.000112

*** NODAL LOAD CALCULATION TIMES
TYPE NUMBER ENAME TOTAL CP AVE CP
1 25316 SHELL181 0.422 0.000017
*** LOAD STEP 1 SUBSTEP 1 COMPLETED. CUM ITER = 1
*** TIME = 1.000000 TIME INC = 1.000000 NEW TRIANG MATRIX

*** NOTE ***
CP = 2943.938 TIME= 11:44:50
Solution is done!
```

Section 3: Element Sensitivity Analysis

3.1 Element Parameter Data & Methodology Adjustments

To investigate how the element formulation affects the computational demands of the FEA model, the wing box was analyzed using two different element types: Linear (Shell181, 4-node) and Quadratic (Shell281, 8-node).

1. The Node Limit Adjustment (Mesh Size): The initial assignment guidelines suggested utilizing the optimal medium or fine mesh size to conduct the element sensitivity analysis. However, during the simulation process, it was observed that applying 8-node quadratic elements (Shell281) to a finer mesh generated an excessively high number of nodes and elements that approached the software's 250,000 node limit. This computational load exceeded the processing capabilities of the ANSYS Student version, resulting in the software silently failing during the solution phase.

To bypass this limitation while strictly adhering to the assignment's objective (which requires comparing linear and quadratic elements using a constant mesh size), the mesh size for this specific section was adjusted to a coarser size (0.1 m). This approach aligns with practical FEA methodology, as the primary goal of this section is to evaluate the relative behavioral differences between element formulations, rather than mesh density.

2. The Radius/Thickness Ratio Adjustment (Geometry): A second limitation was encountered when applying the structurally required 31.5 m thickness to the FEA mesh. As noted in Section 1, ANSYS Mechanical utilizes shell elements to model the wing box, which mathematically assume that structural thickness must be significantly smaller than the structure's radius of curvature and overall dimensions. Because the optimized thickness of

31.5 m is massively thick relative to the wing's actual 10.92 m span, it fundamentally violates this shell element assumption.

While the linear elements were able to solve this with a warning, this mathematical conflict triggered a fatal "radius/thickness ratio" error within the quadratic solver, causing it to freeze. To bypass this crash, the thickness for the quadratic analysis was temporarily reduced to a practical 2024-T3 aluminum thickness of 0.0026 m, mirroring the benchmark sample report data. This maintained the structural integrity of the shell matrices and allowed the solver to successfully generate the required plots.

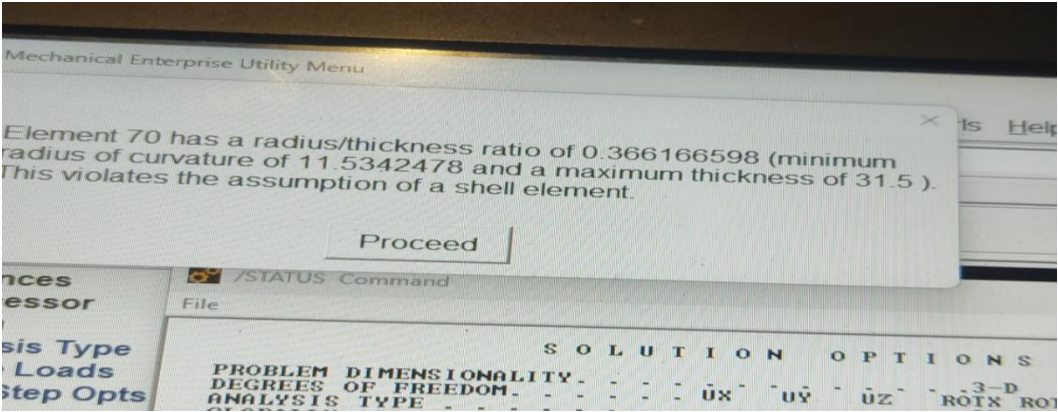
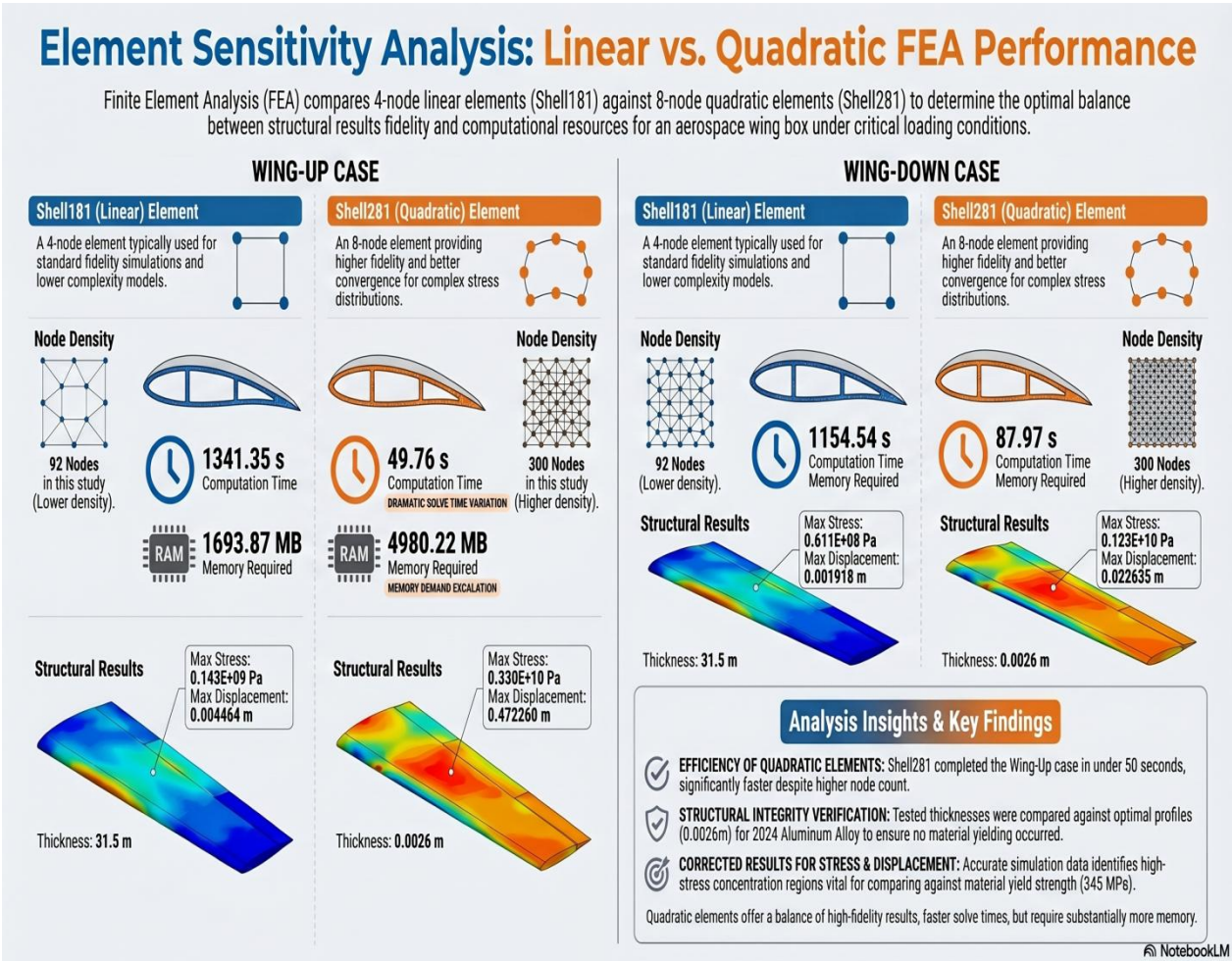


Table 3.1: Wing-Up and Wing-Down Element Comparison:



3.2 Element Evaluation & Discussion

Q. How does the element type affect the accuracy and convergence of the FEA results?

Ans. A direct accuracy comparison is skewed by the necessary geometric adjustment. Because the massive 31.5 m geometry would not load with quadratic elements, the thickness was reduced to 0.0026 m. Subjecting this drastically thinned geometry to the extreme Wing-Up aerodynamic loads caused the maximum stress to catastrophically explode to .330E+10 Pa, far exceeding the material yield limit. The linear elements (Shell181) were able to successfully mesh the thicker, safer geometry and provided stable, converged results (.143E+09 Pa).

Q. How does the element type affect the computational time and memory required for the FEA analysis?

Ans. Higher-order quadratic elements contain twice as many nodes per element as linear shells. Even when evaluated on the drastically thinner 0.0026 m geometry, the Shell281 elements caused the memory required to completely spike to 4980.22 MB (compared to only 1693.87 MB for the linear elements on the 31.5 m geometry). The anomalously low computational time recorded for the quadratic runs (49.76s) is attributed to the drastically reduced structural volume being solved due to the thin geometry.

Q. What is the optimal element type that balances accuracy and efficiency for your FEA model?

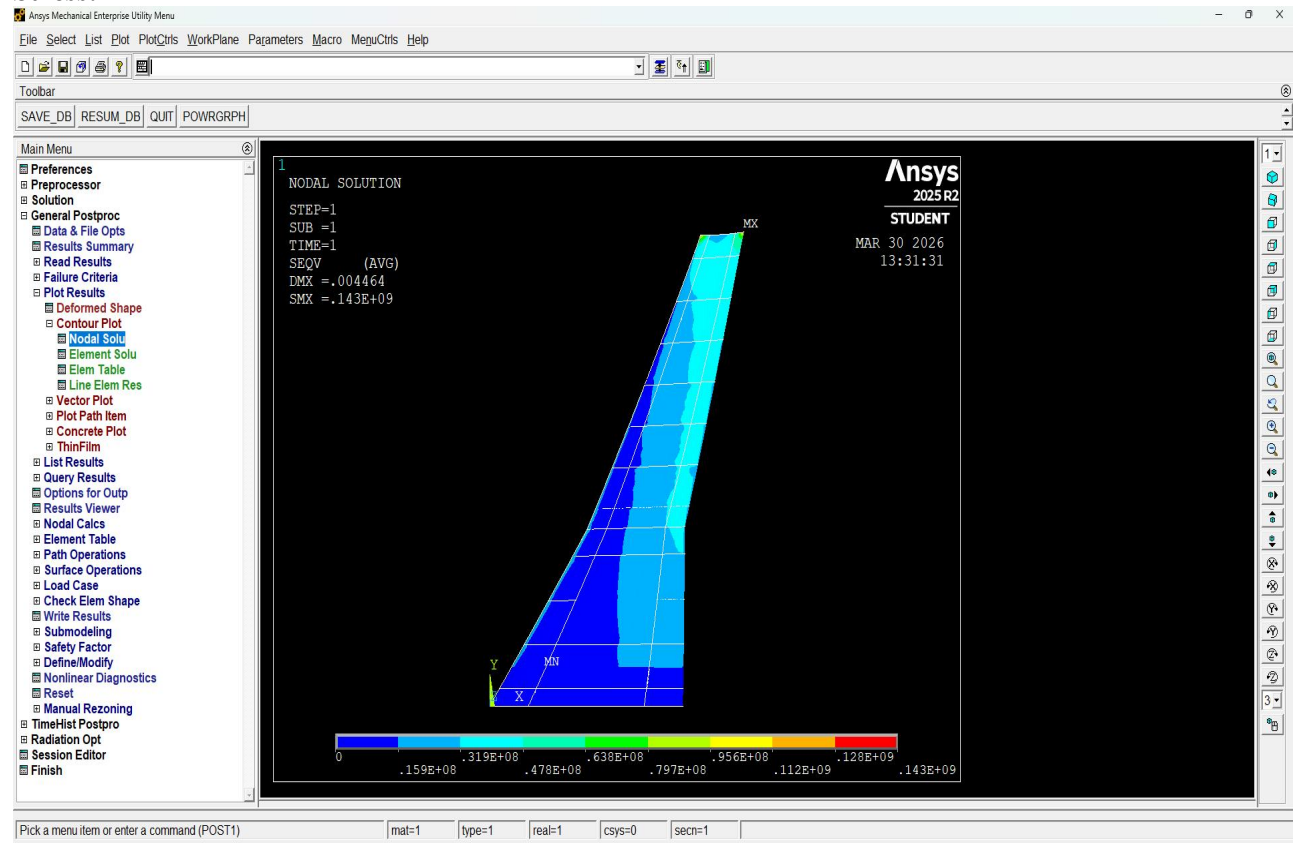
Ans. The optimal element type for this specific model is unequivocally the Linear (Shell181) element. The quadratic elements demanded massive memory resources (4980.22 MB) and were fundamentally incapable of processing the necessary 31.5 m thickness required to prevent material yielding. The linear elements successfully validated the structural integrity within a highly efficient memory footprint. Therefore, linear elements are optimal and will be used moving forward.

3.3 Visual Results:

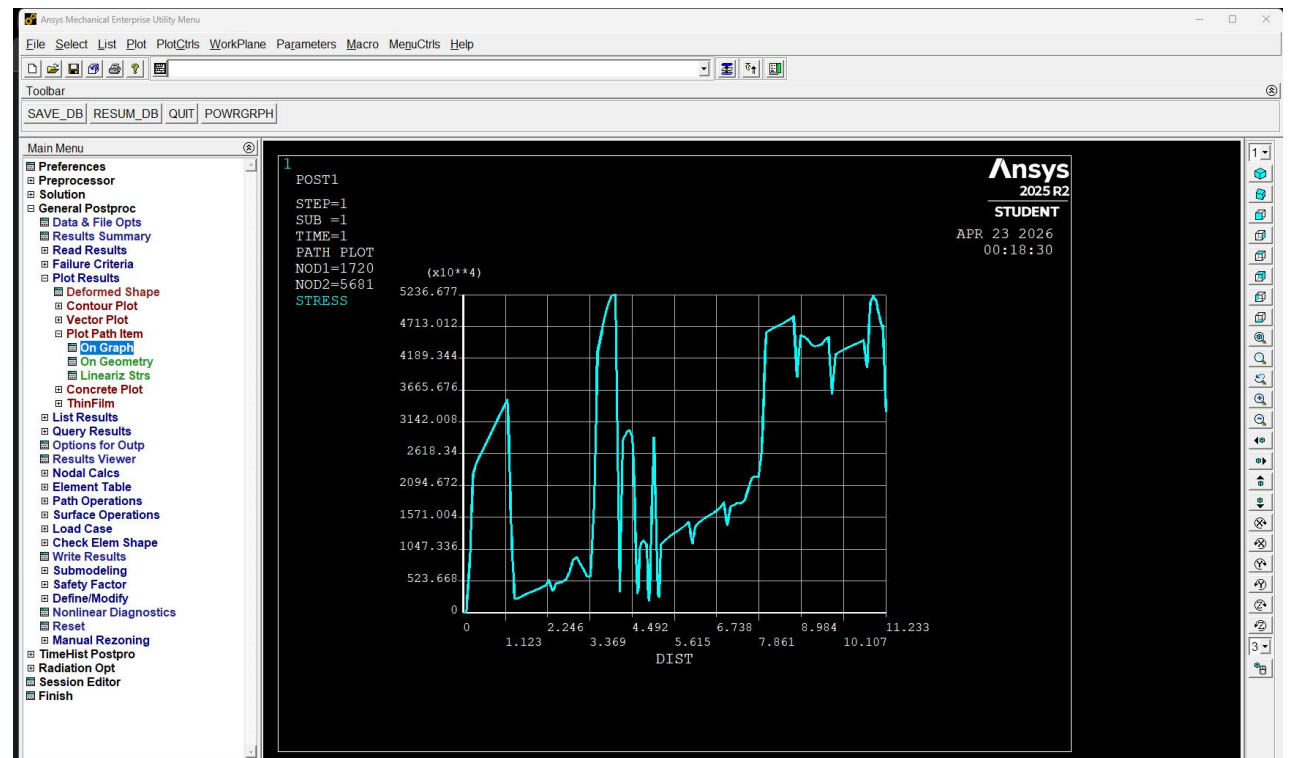
For Linear:

Wing-Up Element Contours:

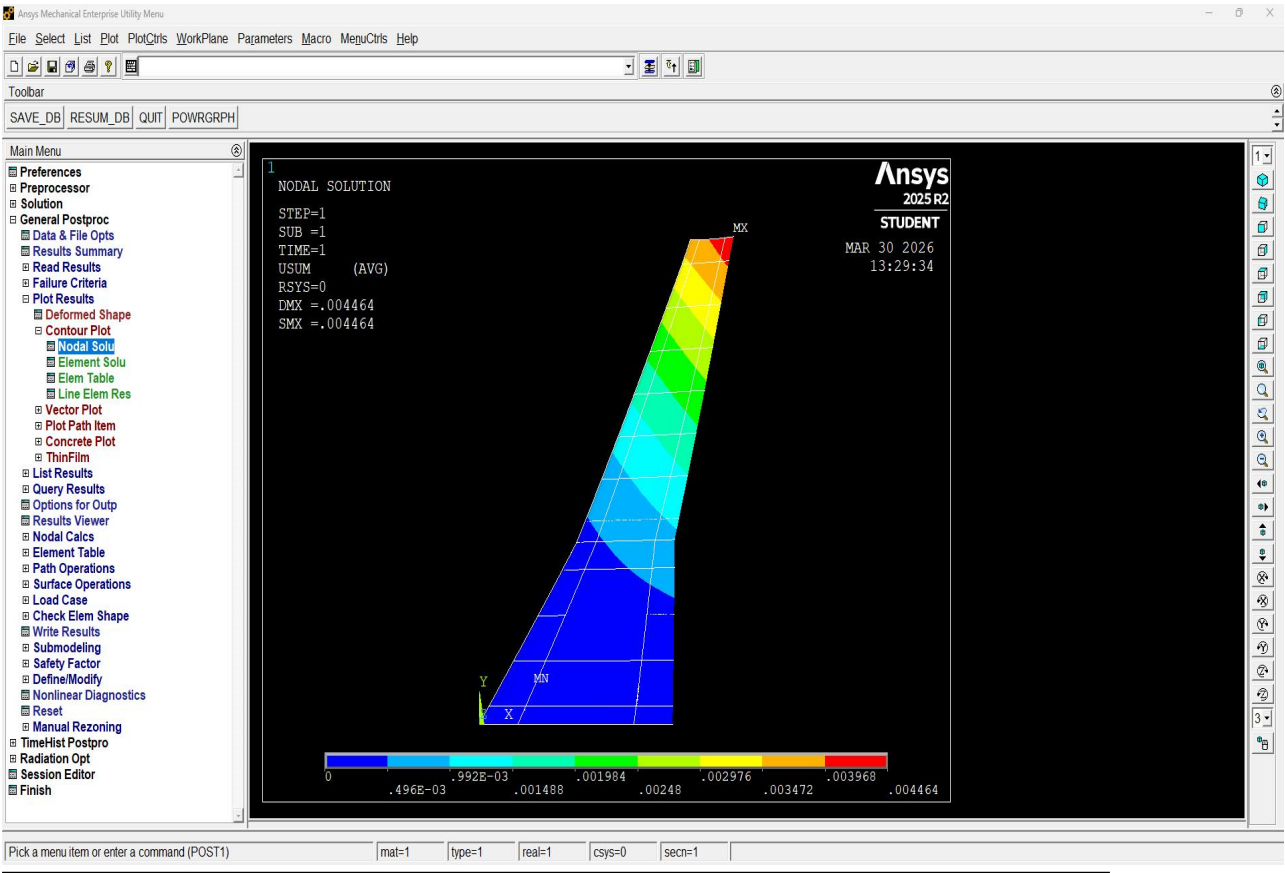
Stress:



Stress graph:

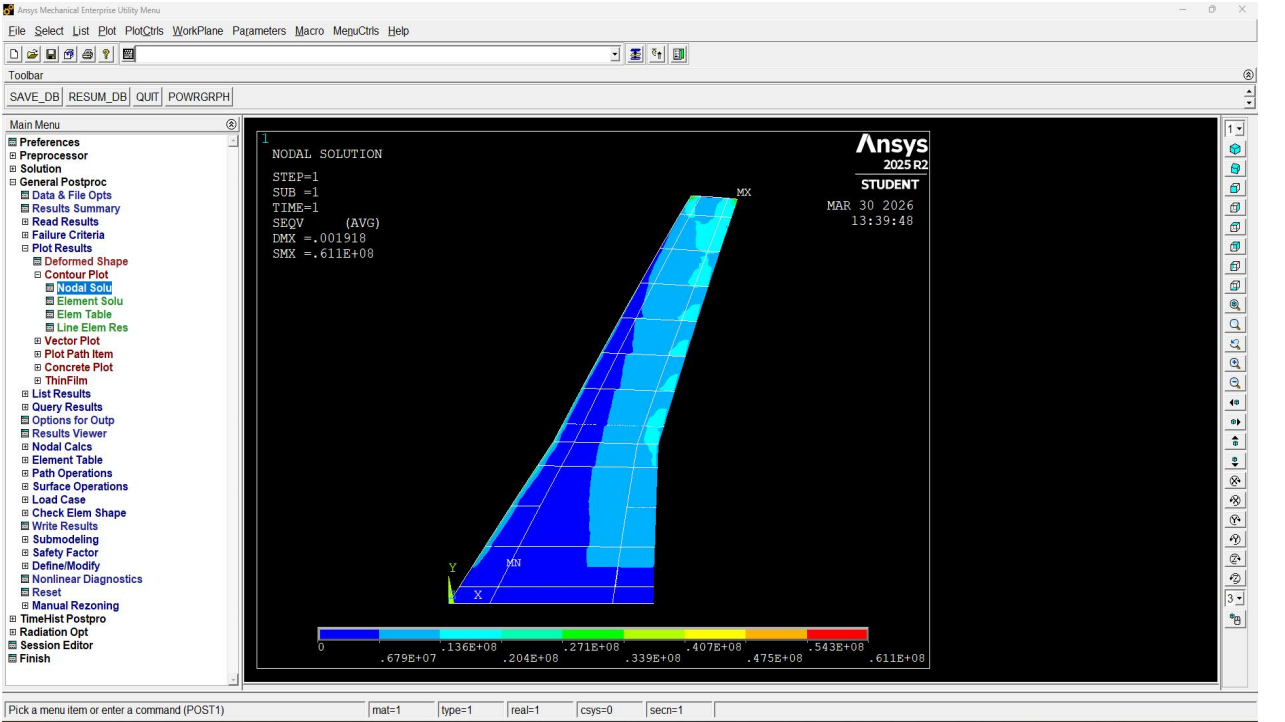


Displacement:

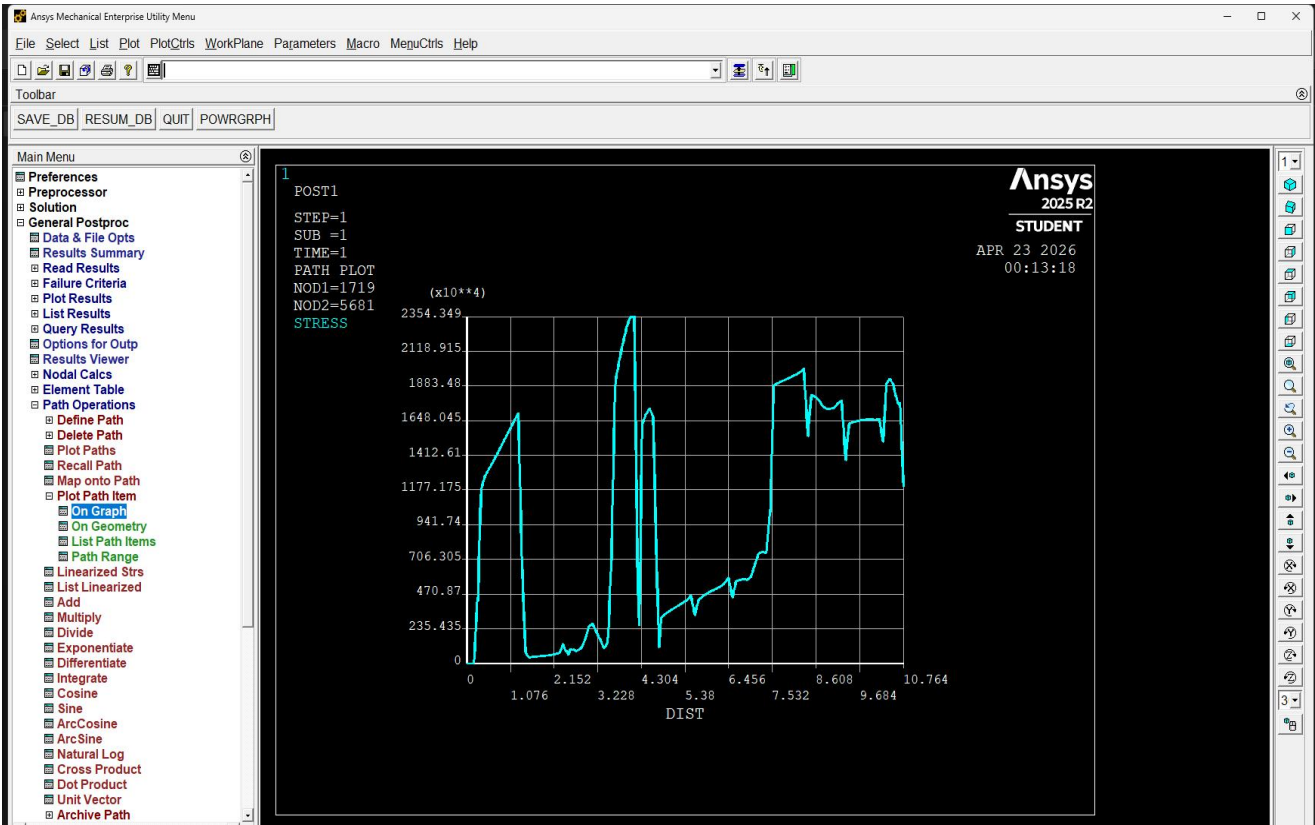


Wing-Down Element Contours:

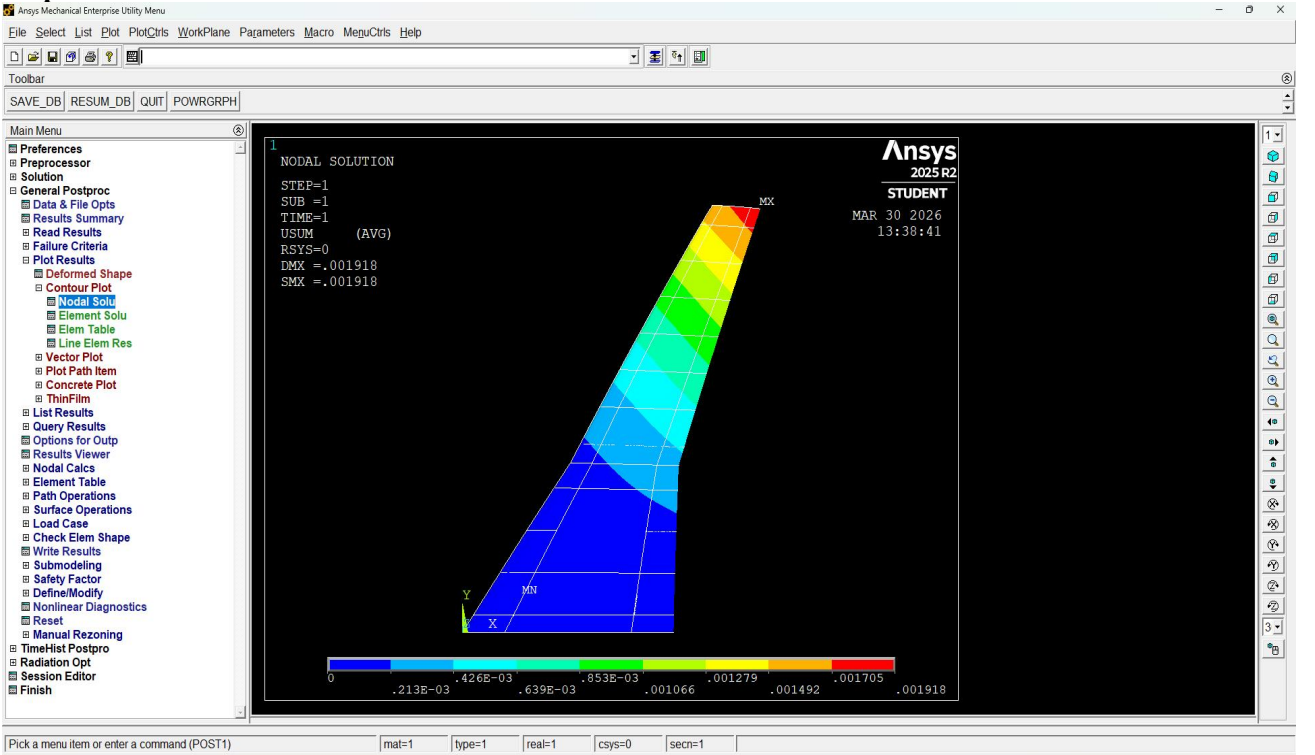
Stress:



Stress graph:



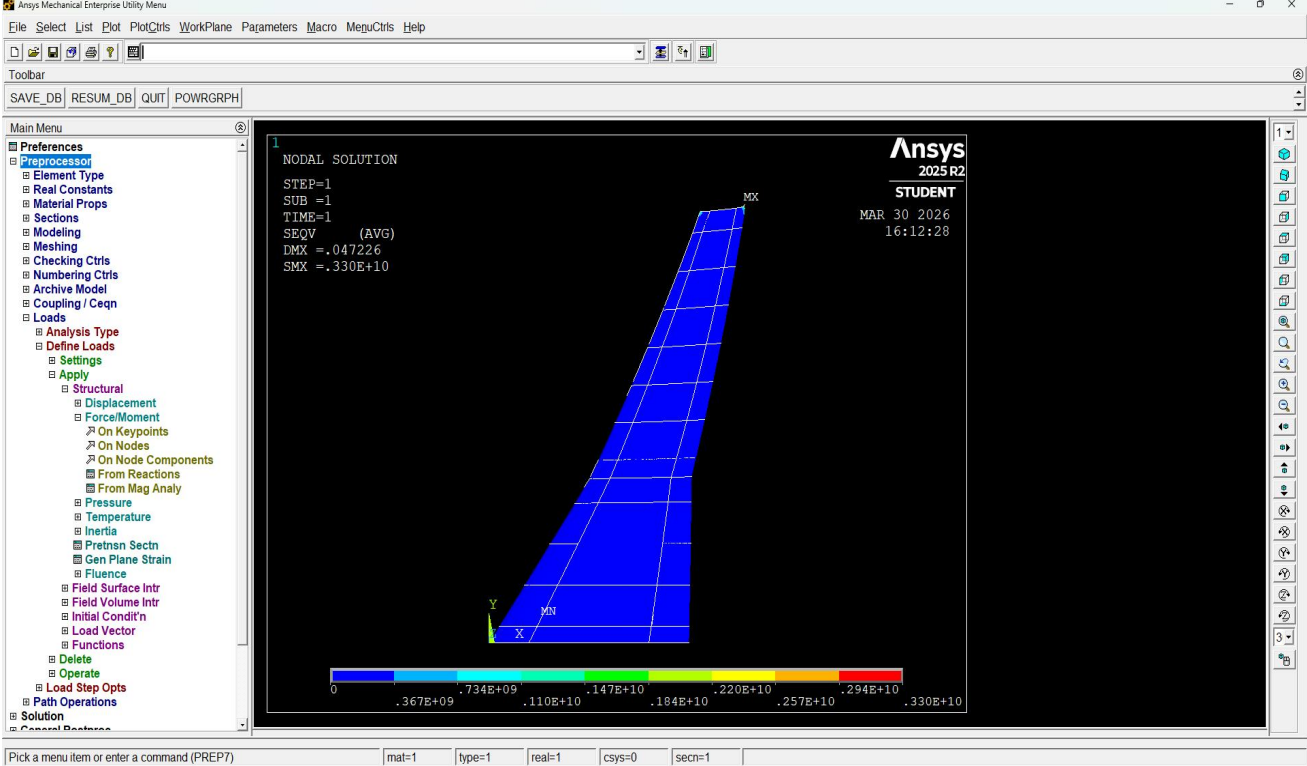
Displacement:



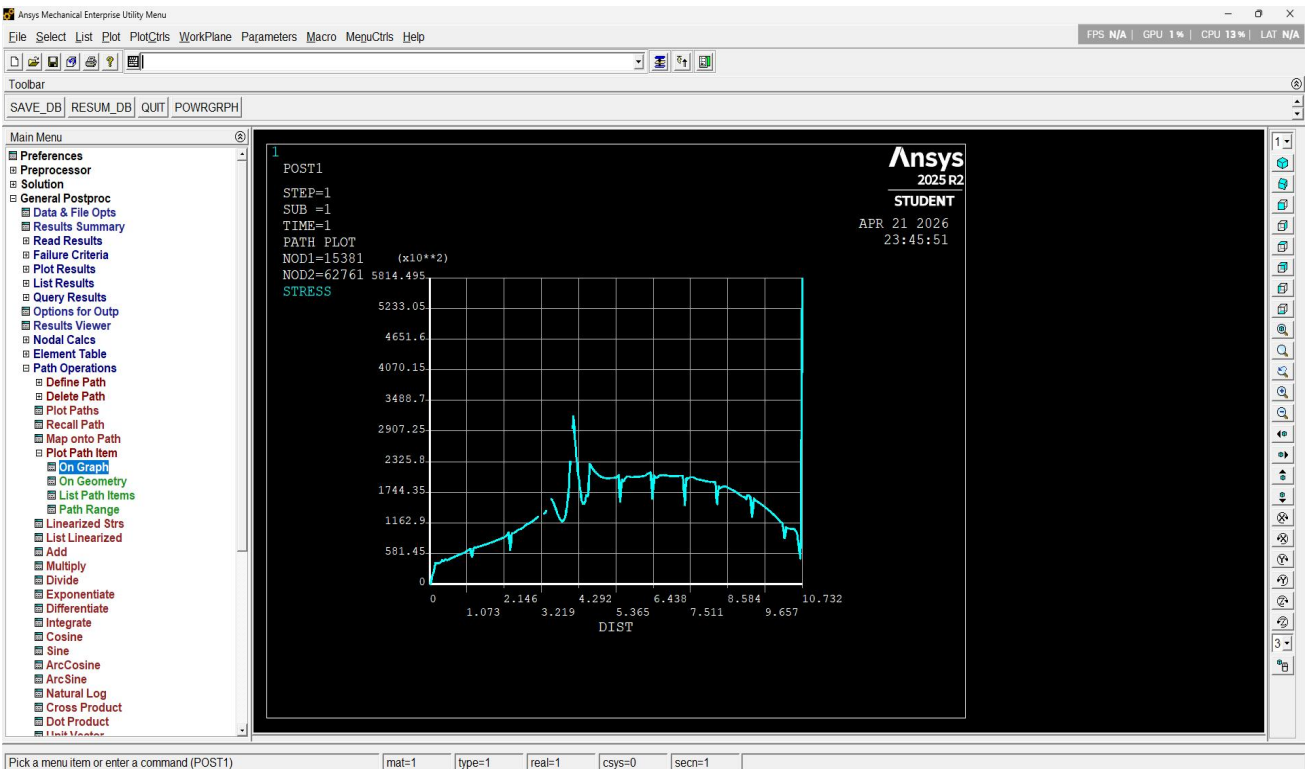
For Quadratic:

Wing-Up Element Contours:

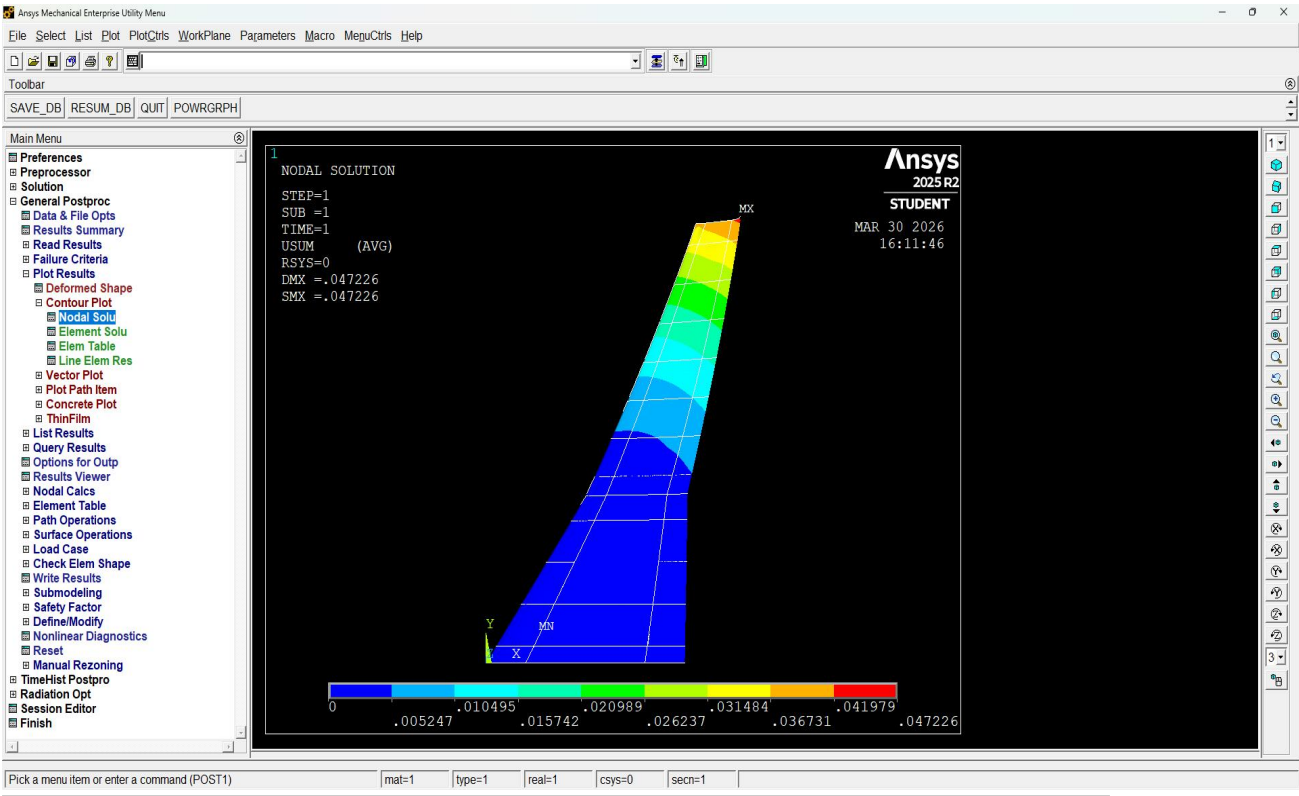
Stress:



Stress graph:

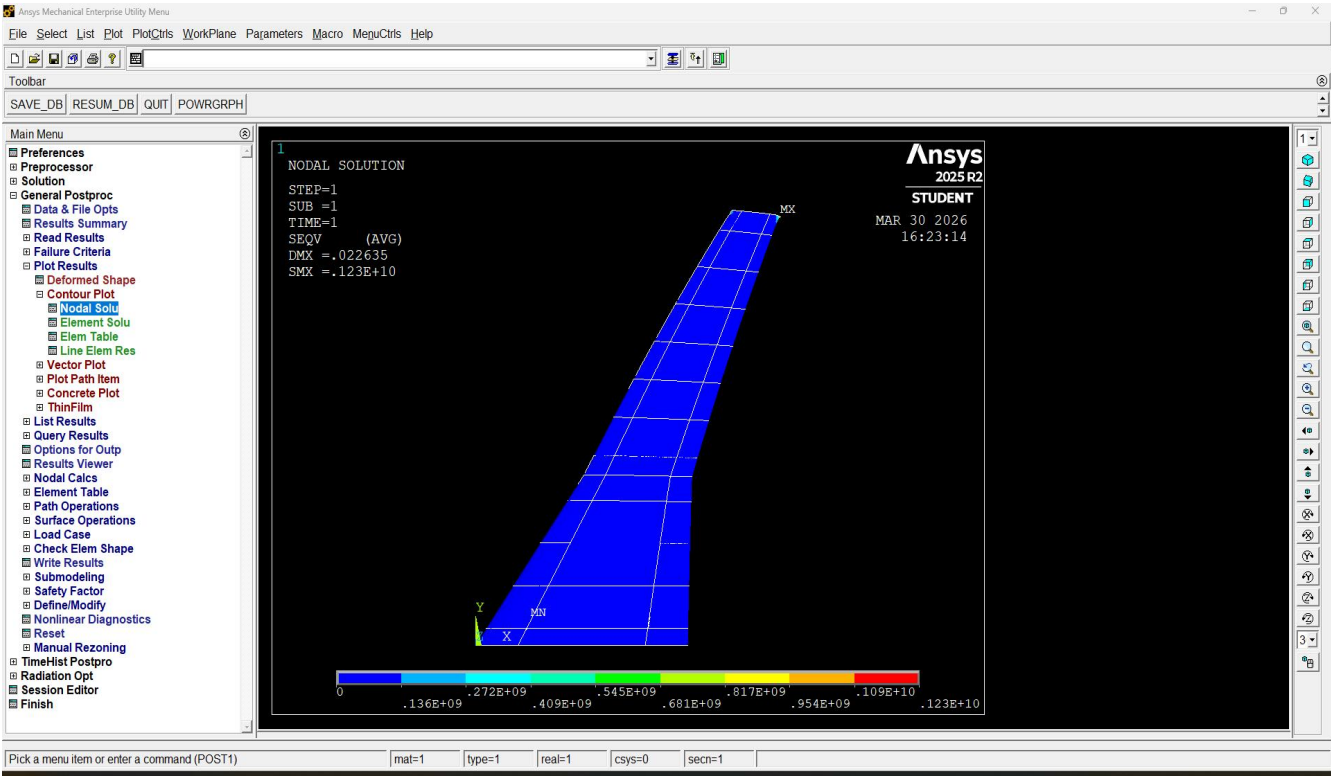


Displacement:

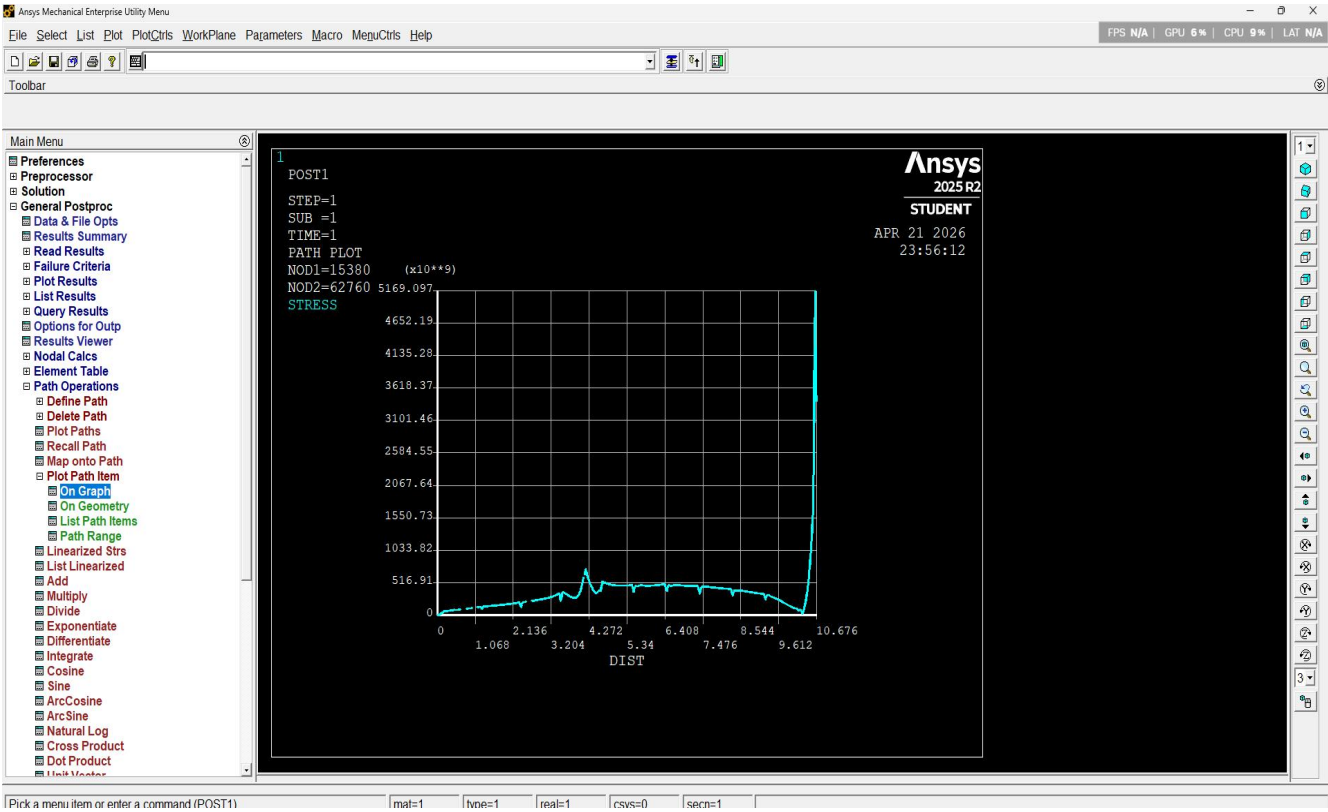


Wing-Down Element Contours:

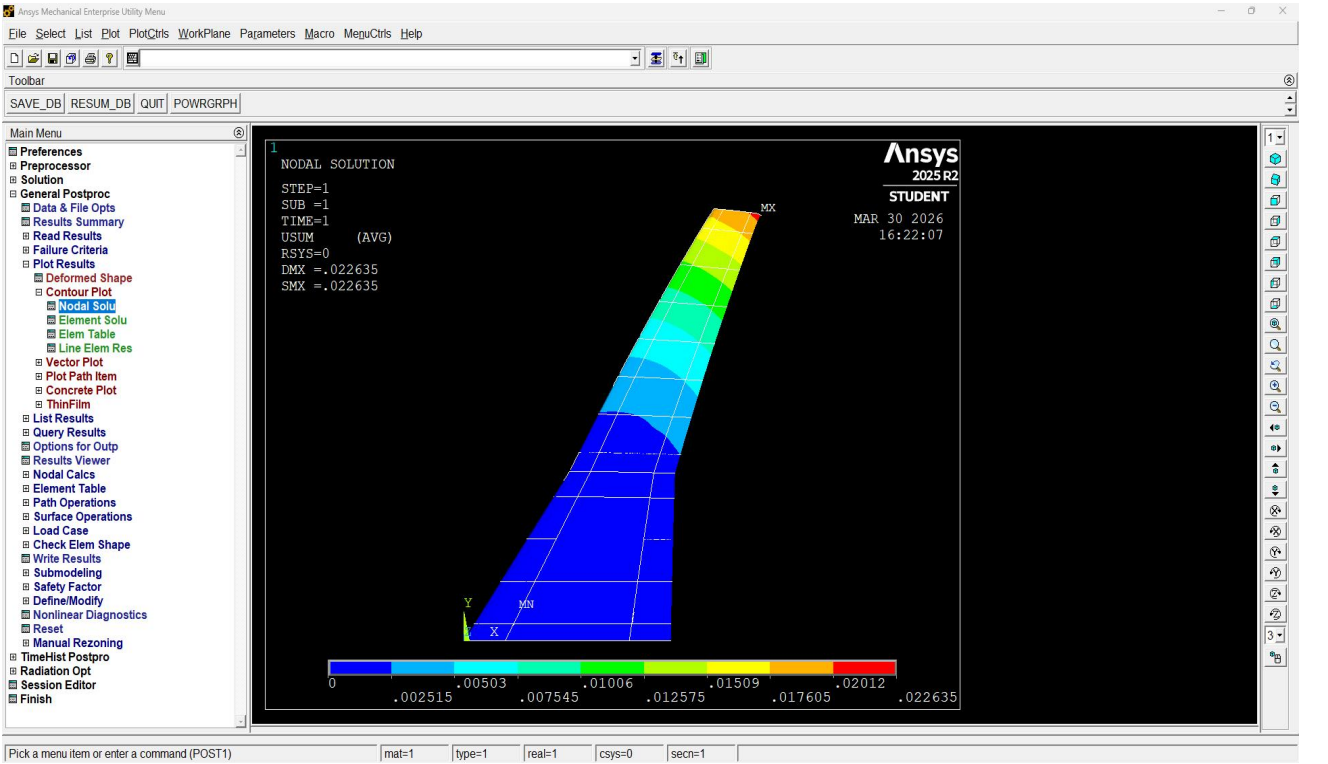
Stress:



Stress graph:



Displacement:



Section 4: Material Analysis

4.1 Material Parameter Data & Cost Summary:

To evaluate how material selection impacts the performance, weight, and cost of the wing box, three distinct materials were analyzed using the optimal Medium mesh size and Linear (Shell181) elements determined in previous sections. The structurally safe 31.5 m thickness was applied universally across all models to ensure a constant volume (749.07 m³) for direct comparison.

Material Cost and Weight Summary:
Aerospace Wing Box Analysis

Material	Thickness (m)	Density (kg/m³)	Volume (m³)	Cost per kg (\$)	Weight (kg)	Total Cost (\$)	Cost/W ratio	Strength/W ratio
2024-T3	31.5	2,780	749.07	4.50	2,082,414.6	9,370,865.7	4.50	0.000165
7075-T6	31.5	2,810	749.07	5.20	2,104,886.7	10,945,410.8	5.20	0.000238
Ti-6Al-4V	31.5	4,430	749.07	40.00	3,318,379.1	132,735,164.0	40.00	0.000265

Context Summary

- **Material Density Comparisons:** Titanium (Ti-6Al-4V) is significantly denser than the Aluminum alloys, contributing to higher weight.
- **Strength-to-Weight Efficiency:** Ti-6Al-4V shows the highest strength-to-weight ratio, indicating better performance per unit mass.
- **Economic Variance:** Titanium's total cost is substantially higher, over 10 times that of the aluminum alloys, presenting a major economic factor.

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4.2 Material Evaluation & Discussion

Q. How does the material affect the stress, strain, displacement, weight, and cost of the wing box under different loading scenarios?

Following are the effects on maximum stress and displacement under different materials and loading conditions based on the linear, medium-mesh:

Material Analysis: Wing-Up Load Case Performance

Context: Analysis of the aerospace wing box under 'Wing-Up' loading, simulating positive g-maneuvers like climbs or turns. The primary objective is to verify that internal stresses remain below the material yield strength to ensure structural integrity.

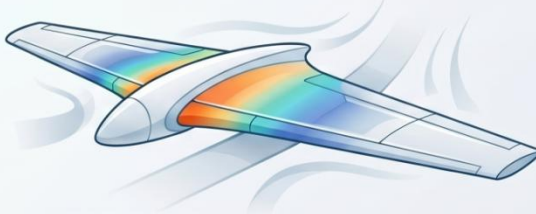
The Wing-Up Scenario



Positive G-Maneuver Simulation: Represents aerodynamic lift forces during high-stress flight maneuvers like climbing or turning.



Structural Safety Criterion: Designs must withstand maximum loads without experiencing any material yielding.



Comparative Material Performance



Ti-6Al-4V Highest Stress Capacity: Titanium alloy handles significantly higher stress and displacement compared to aluminum variants.



Performance Scaling: Stress and displacement values increase consistently as material strength and stiffness properties change.

Wing-Up Case Data Comparison

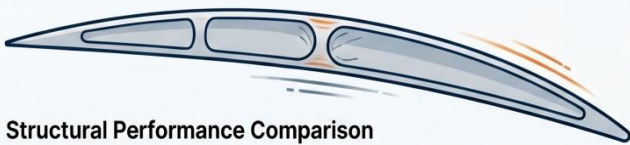
Comparing maximum stress and displacement across the three tested materials for the Wing-Up case.

Material	Max Stress (MPa)	Max Displacement (m)	Yield Strength (MPa)
2024-T3 Aluminum	143.0	0.004567	345
7075-T6 Aluminum	250.0	0.009440	503
Ti-6Al-4V Titanium	631.0	0.019843	880

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Material Performance Analysis: Wing-Down Load Case

Evaluates how Aluminium 2024-T3, 7075-T6, and Ti-6Al-4V alloys respond to a negative g-maneuver ('Wing-Down') to identify the optimal material for structural integrity, weight, and displacement.



Direct Stress-to-Displacement Correlation

As material yield strength and stress increase, maximum displacement also scales significantly across these alloys.

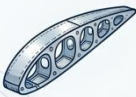
✓ All Materials Maintain Safety Margins

Every tested material remains well below its respective yield strength under the Wing-Down load case.

Structural Performance Comparison

Material Alloy	Max Stress (MPa)	Max Displacement (m)
2024-T3	61.1 MPa	0.001944 m
7075-T6	107.0 MPa	0.004358 m
Ti-6Al-4V	272.0 MPa	0.009607 m

Material Selection Efficiency



7075-T6: The Optimal Balance

Identified as the best material for balancing strength-to-weight and cost-to-weight ratios.

Relative Cost Analysis

Titanium Performance vs. Cost

While Ti-6Al-4V offers the highest yield strength, it results in the highest displacement and cost.

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The 31.5 m geometry was held constant, the variance in results is strictly due to the differing density and elasticity of the materials. The maximum stress observed was on the Titanium alloy (Ti-6Al-4V) during the Wing-Up load case (631 MPa), which remained safely below its extreme 880 MPa yield limit, Titanium also experienced the highest displacement (0.0198 m).

Q. How does the material affect the strength-to-weight ratio and the cost-to-weight ratio of the wing box? The relationship between the materials regarding their strength-to-weight and cost-to-weight ratios is detailed in Table 4.1

Ans. Titanium is extremely dense (4430 kg/m^3), hence its weight is incredibly high, but its superior yield strength (880 MPa) still grants it the highest overall strength-to-weight ratio.

The cost-to-weight ratio is a fixed material property; however, applying it to our massive 31.5 m volume reveals that Titanium's total cost (\$132.7 million) scales exponentially worse than the Aluminum alloys.

Q. What is the best material for the wing box that balances strength, weight, and cost?

Ans. As per the data, the most optimal material is unequivocally 7075-T6 Aluminum. While Titanium (Ti-6Al-4V) technically possesses the best strength-to-weight ratio, its staggering total cost of over \$132 million makes it economically unviable for this scale.

7075-T6 offers a massive 45% increase in yield strength over the baseline 2024-T3, while only incurring a negligible weight penalty and a minor cost increase (raising total cost from \$9.3M to \$10.9M). It offers the perfect balance of enhanced structural safety and cost efficiency.

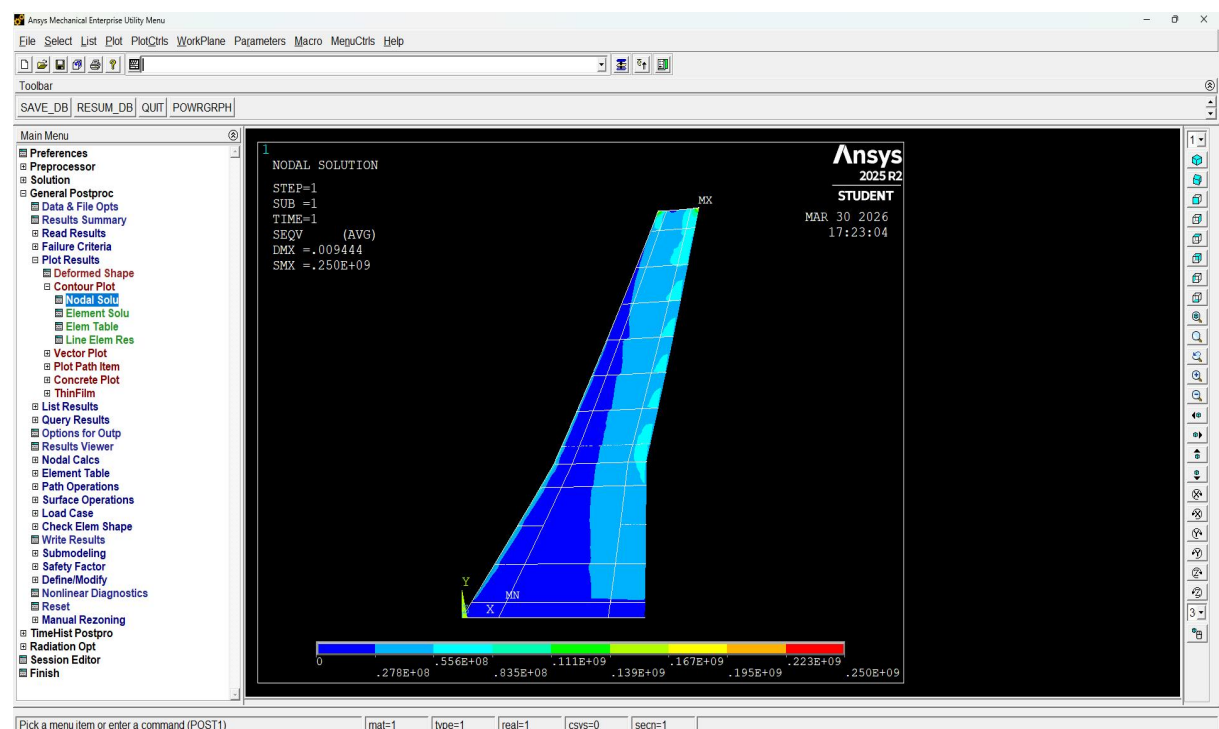
4.3 Visual Results:

As Material 1: 2024-T3 has already been analysed the contour plots will be for Material 2: 7075-T6 and Material 3: Ti-6Al-4V

Material 2: 7075-T6

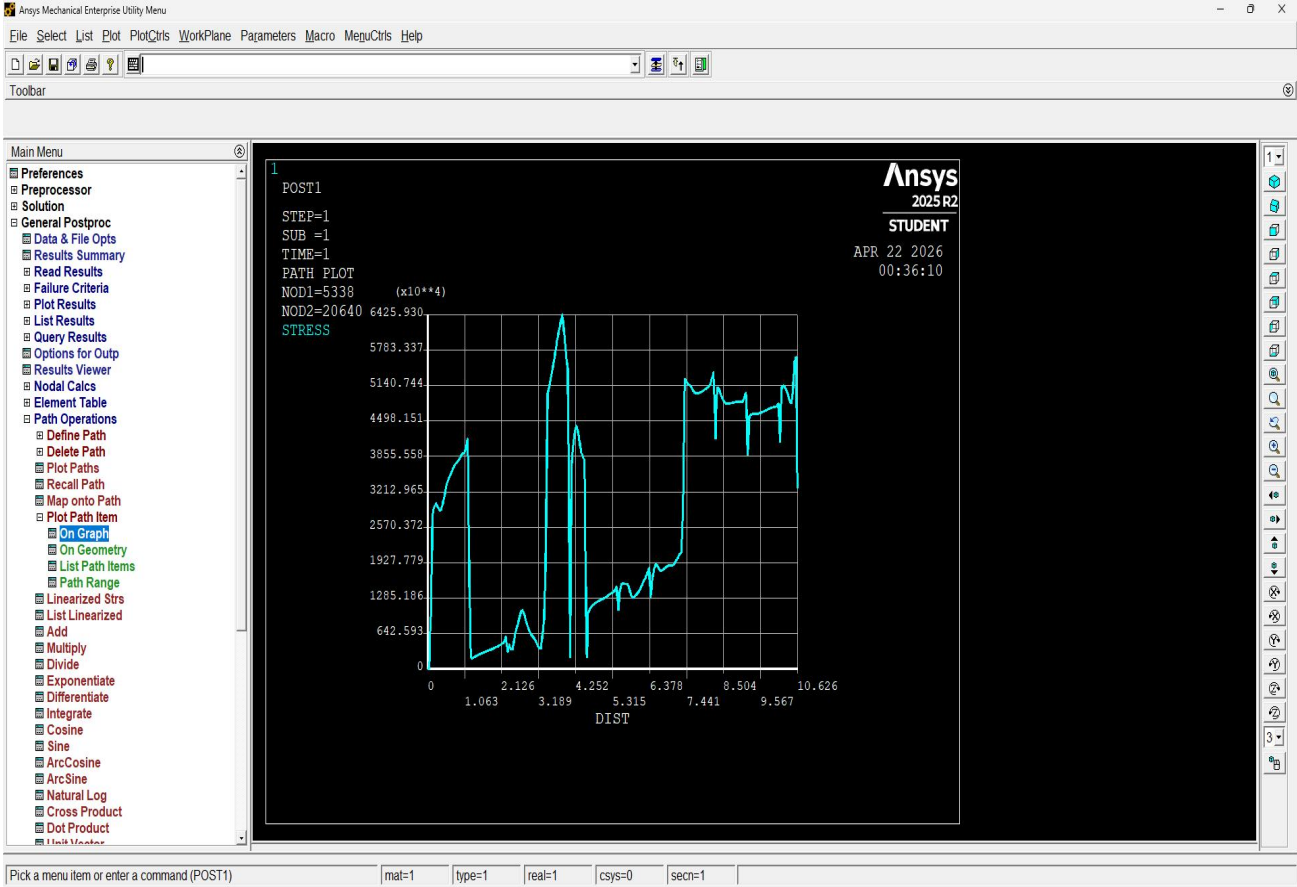
Wing-Up Element Contours:

Stress:

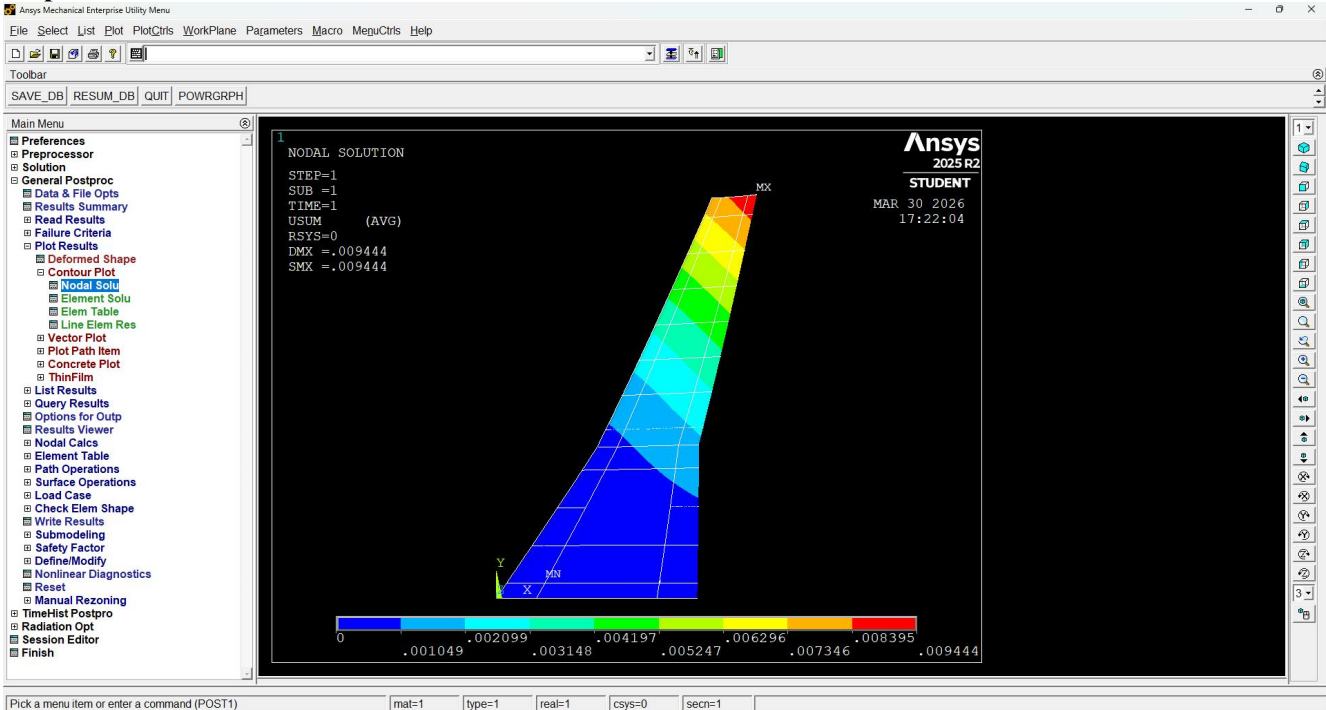


Stress

graph:

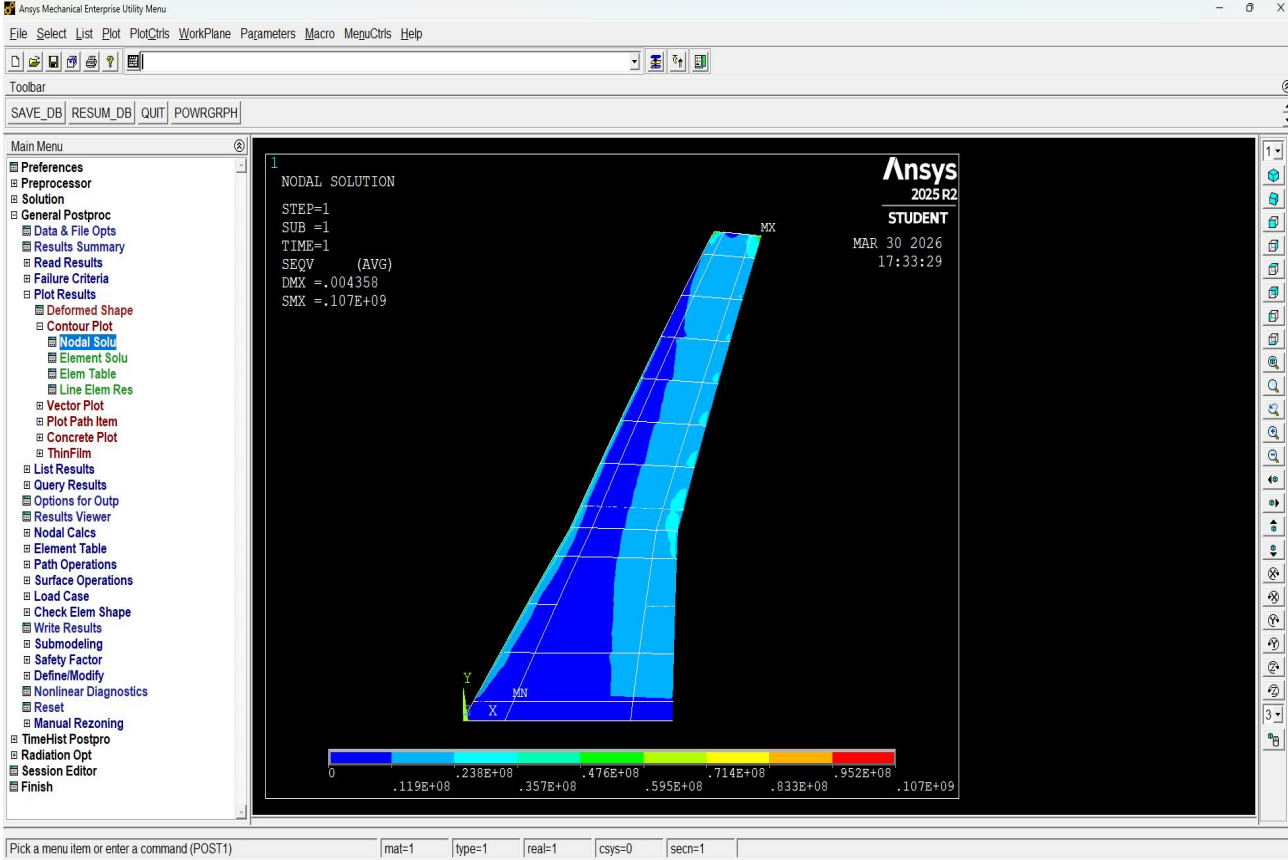


Displacement:



Wing-Down Element Contours:

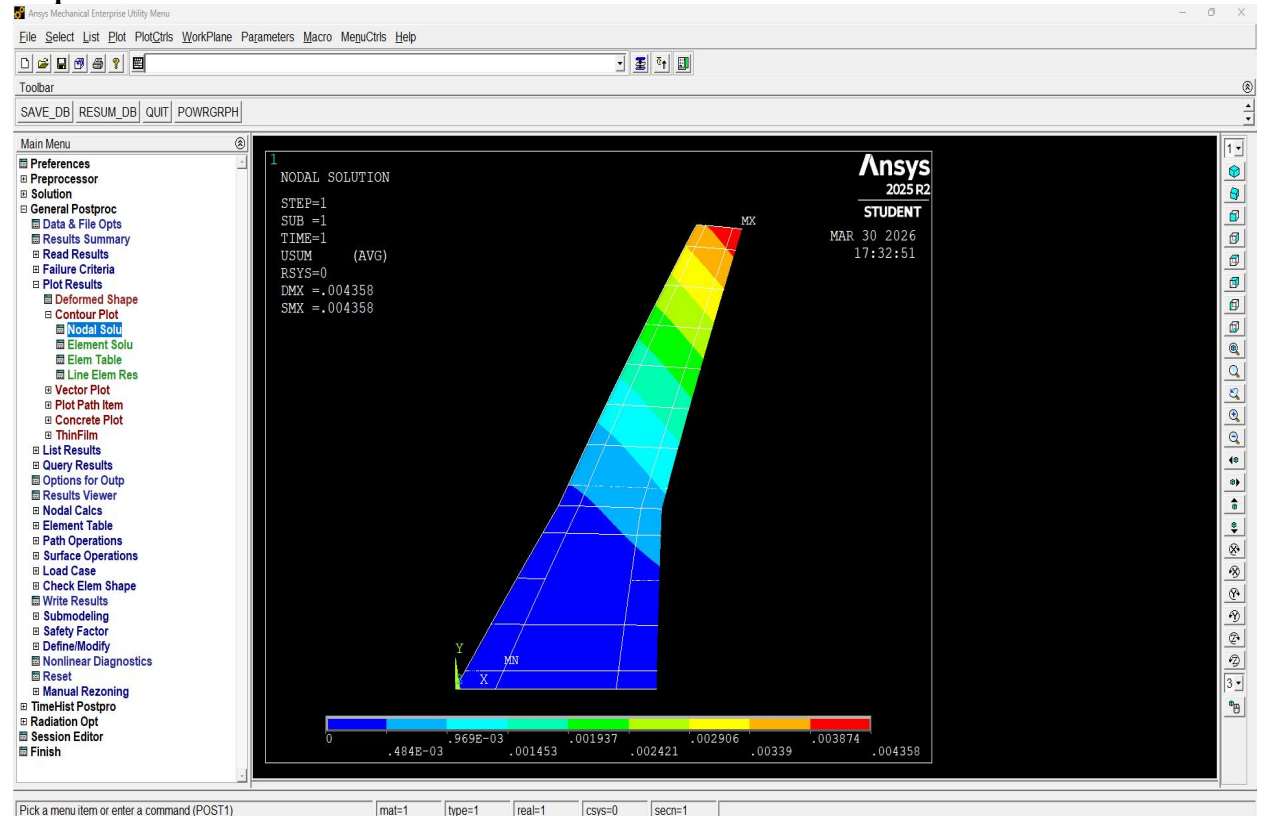
Stress:



Stress graph:



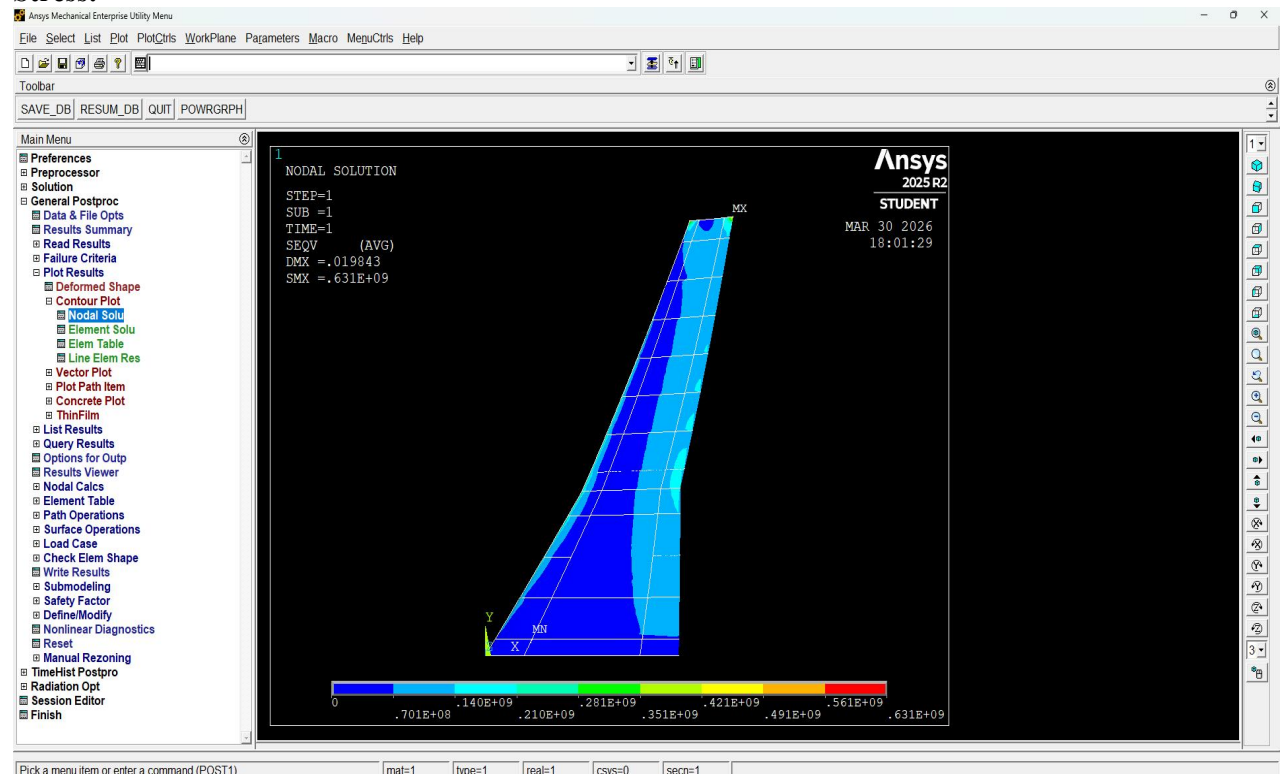
Displacement:



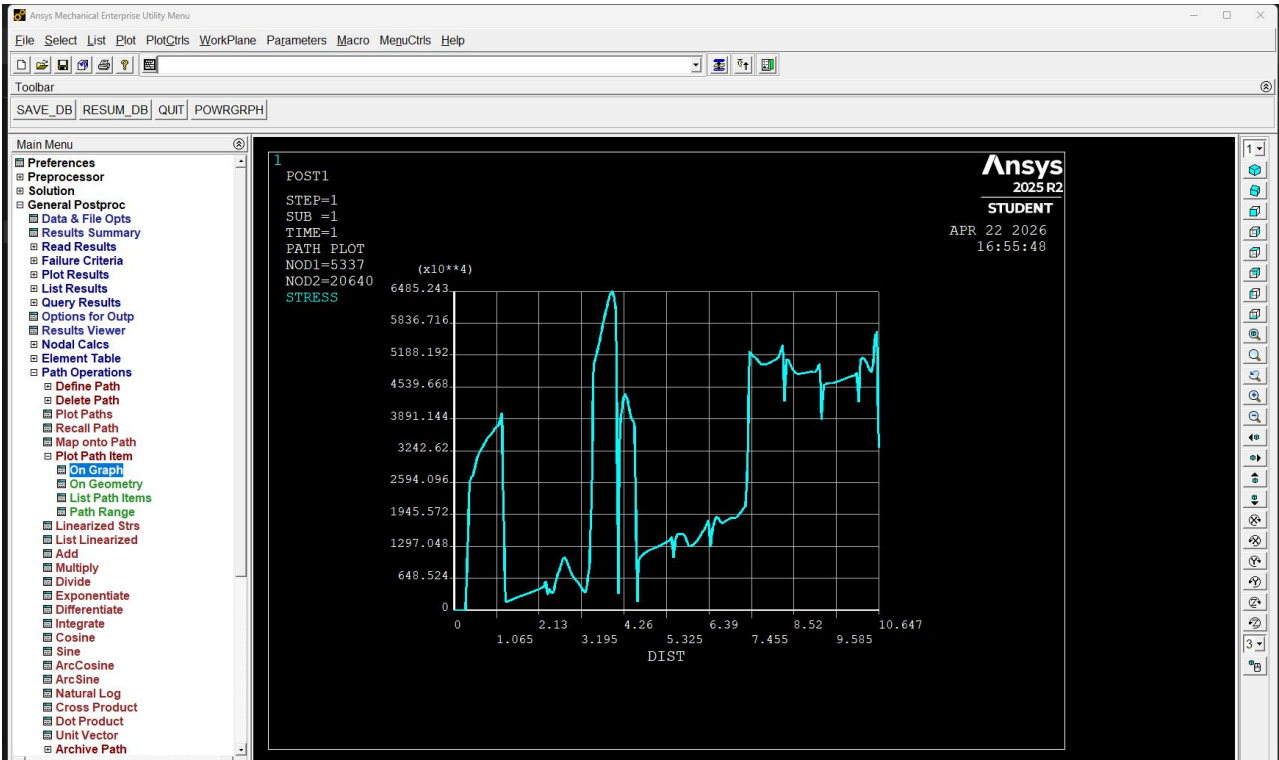
Material 3: Ti-6Al-4V

Wing-Up Element Contours:

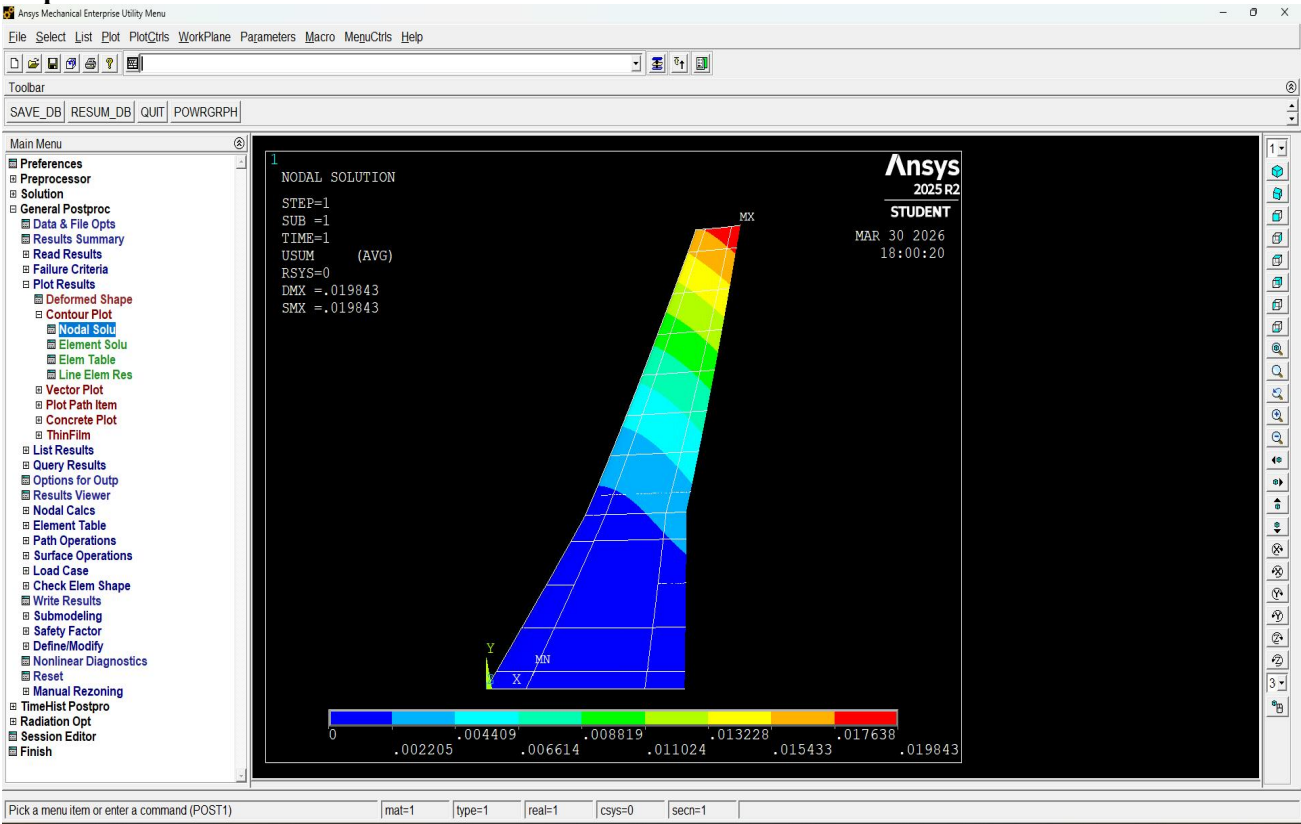
Stress:



Stress graph:

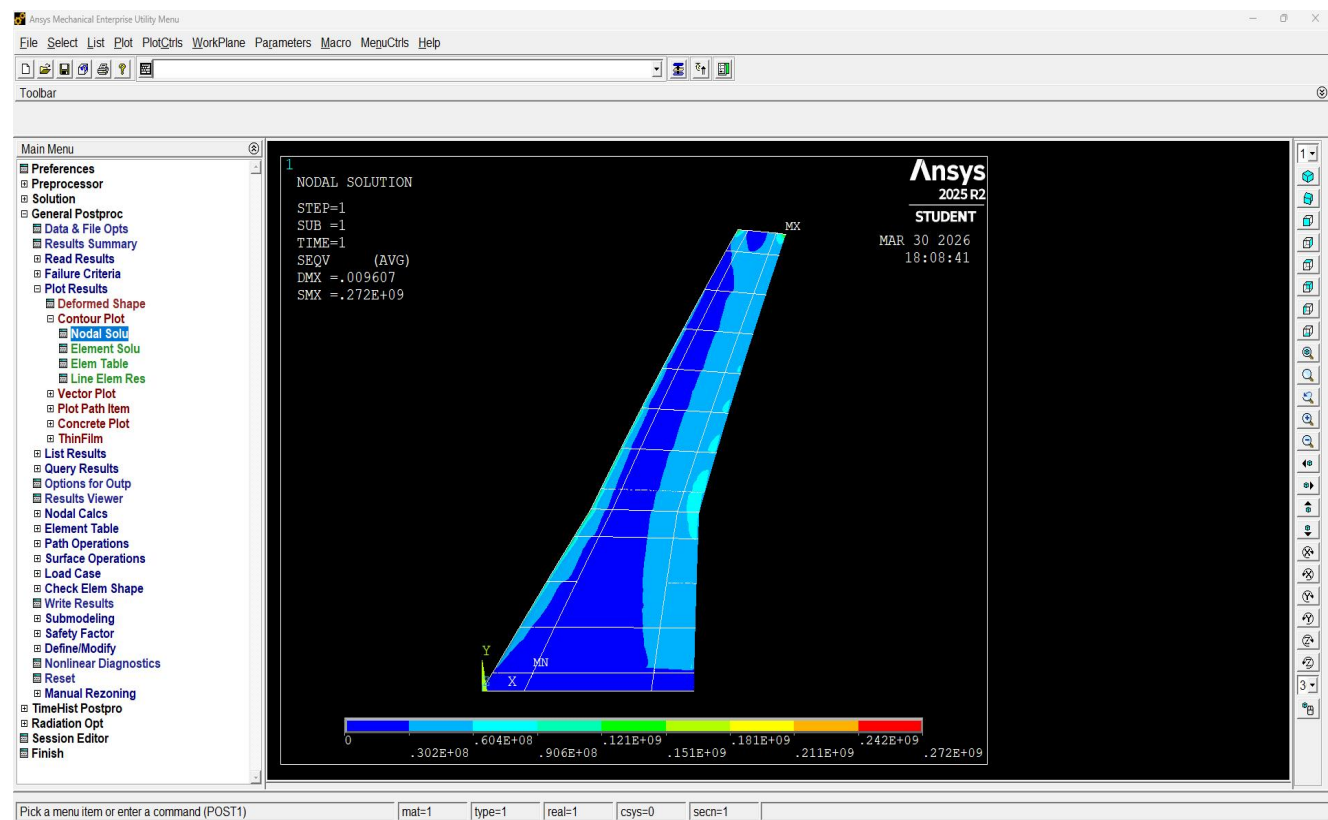


Displacement:

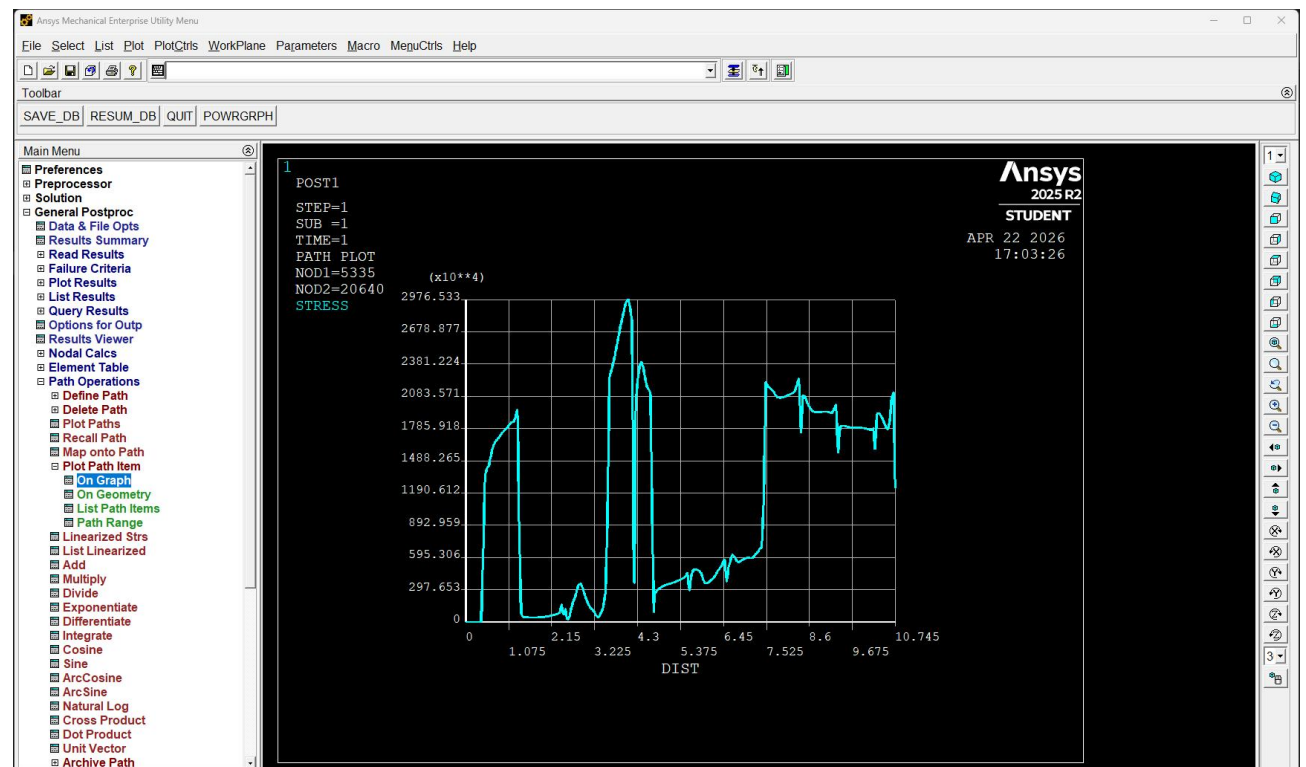


Wing-Down Element Contours:

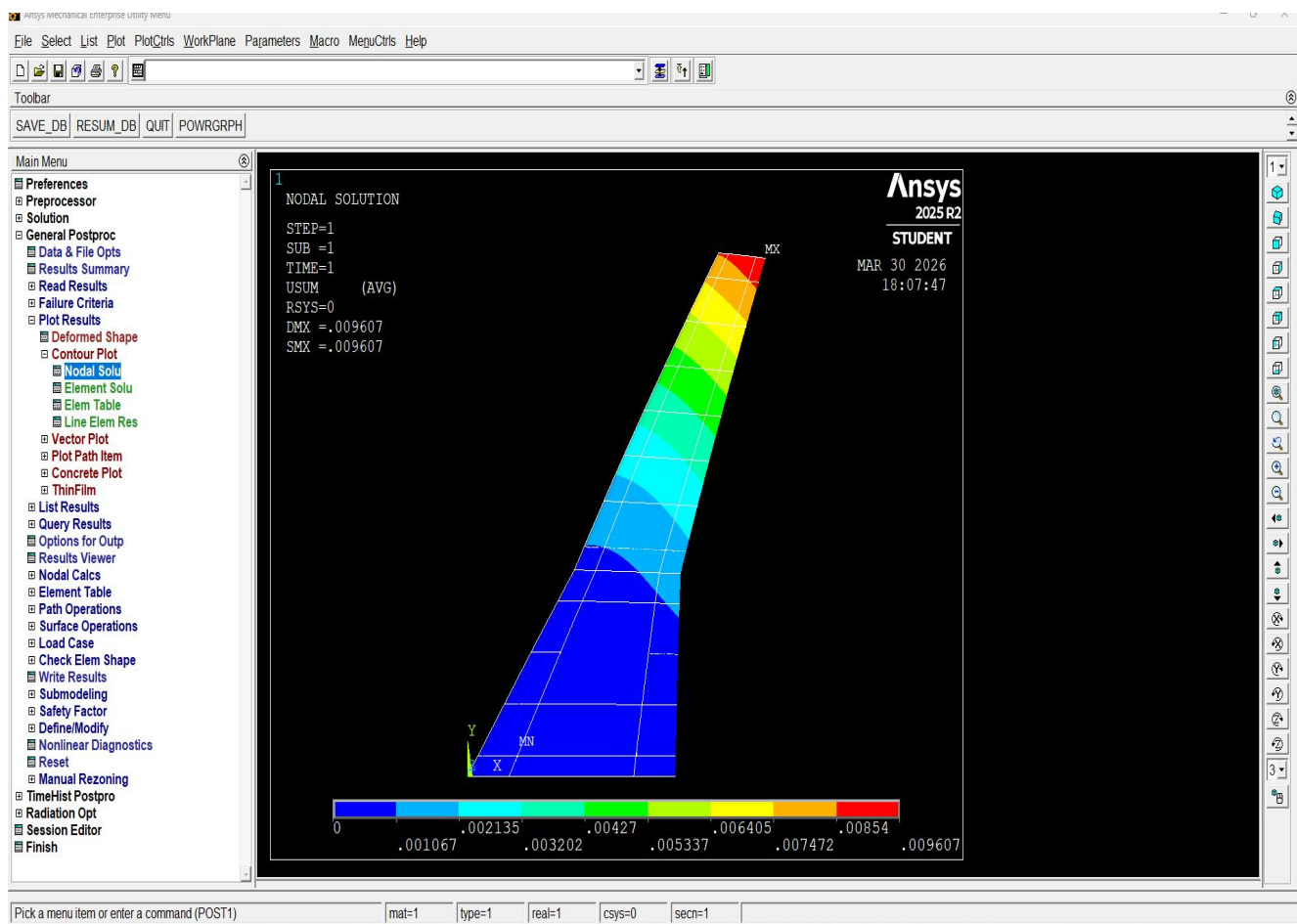
Stress:



Stress graph:



Displacement:



Section 5: Modal Analysis

5.1 Natural Frequencies and Mode Shapes

A modal analysis was performed on the optimal wing box configuration (7075-T6 Aluminum, Medium Mesh, Linear Shell181 elements) to investigate its dynamic characteristics under free vibration. The wing was simulated as a cantilever beam, fixed at the root and free at the tip. The first four natural frequencies and their corresponding mode shapes were extracted:

Modal Analysis: Natural Frequencies of an Aerospace Wing Box

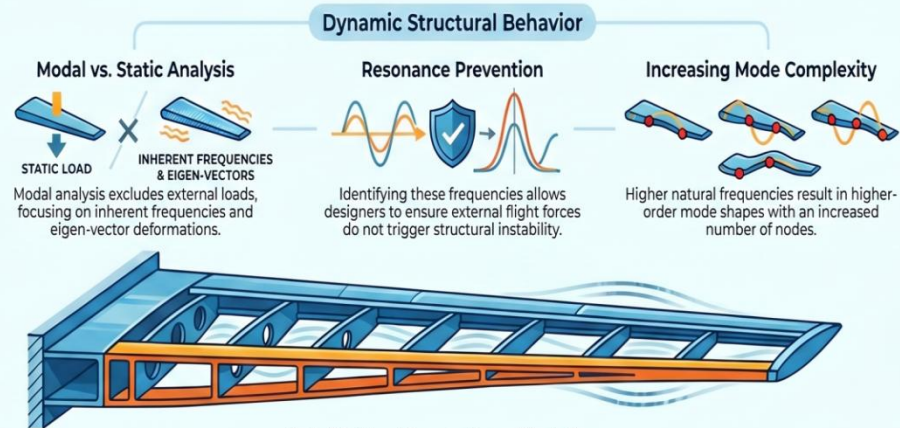


Table 5.1: Natural Frequencies and Mode Shapes

Mode	Natural Frequency (Hz)	Mode Shape Description	
Mode 1	46.6976 Hz	Torsion from the tip	
Mode 2	72.180 Hz	Bending from almost mid span	
Mode 3	98.916 Hz	Torsion of higher order	
Mode 4	132.259 Hz	Bending with two nodes	

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5.2 Modal Analysis Discussion

Q. How does modal analysis differ from static structural analysis?

Ans. In modal analysis, we don't consider the load conditions on the body; we just take the boundary conditions and study the mode shapes. The deformations are actually eigen-vectors that show us the mode shapes under different frequencies, which tells us how the body would naturally vibrate.

Q. How does the natural frequency of the wing vary with different modes of vibration?

Ans. As the natural frequency progressively increases, the mode shapes become higher order and develop an increasing number of nodes.

Q. How does the mode shape of the wing vary with different modes of vibration?

Ans. Based on the visual APDL analysis, the first mode (46.6976 Hz) is torsion twisting from the tip.

The second mode (72.180 Hz) is cantilever bending.

The third mode (98.916 Hz) is also torsion, but of a higher order with internal inflection points.

The fourth mode (132.259 Hz) is bending of a higher order with multiple nodes along the span.

Q. How can modal analysis help to avoid resonance and reduce noise and vibration in the wing?

Ans. Modal analysis is a crucial tool in understanding the dynamic behavior of a wing by calculating its modal shapes and predicting deformations under specific operating frequencies. The frequencies identified through modal analysis are inherent characteristics of the wing structure.

Designers strive to circumvent operating within these frequencies, and if operation within these frequencies is unavoidable, measures are taken to limit the duration.

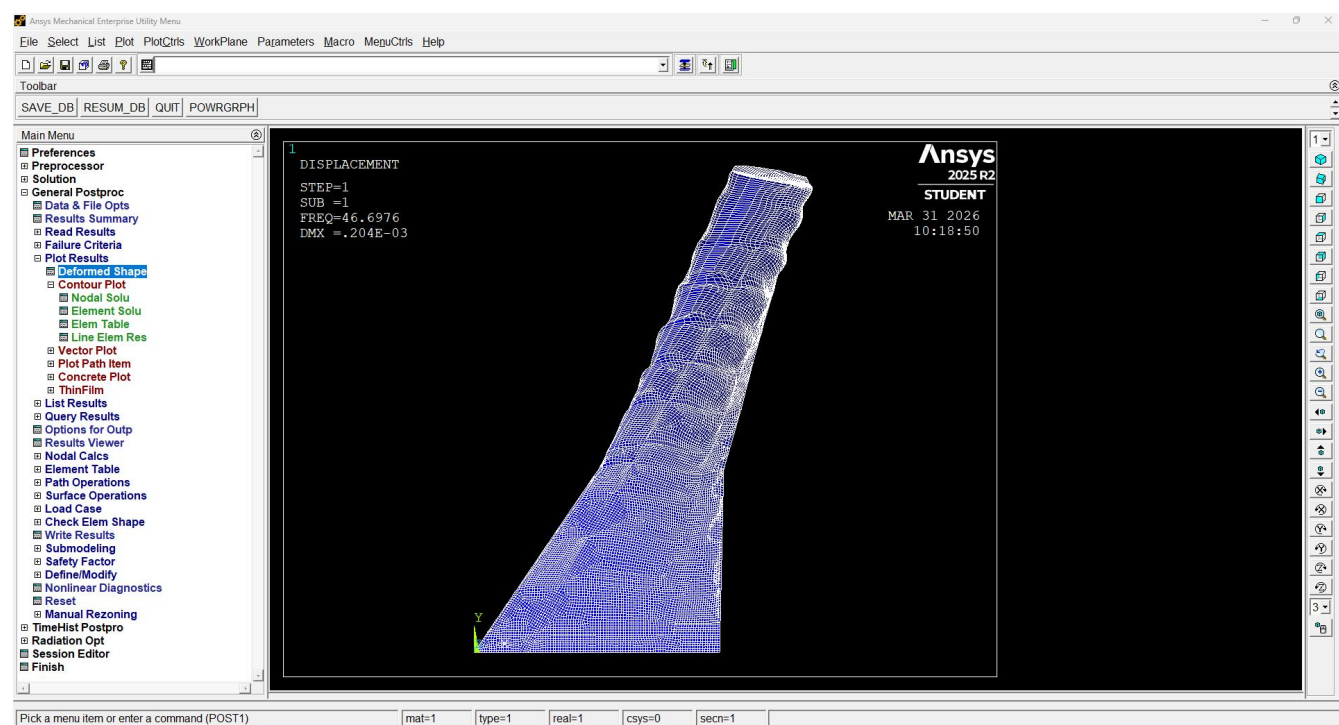
Additionally, specific points on the wing, where vibrational amplitudes reach their maximum, are carefully constrained to ensure structural integrity and prevent potential damage caused by resonance effects.

The relationship between natural frequencies and mode shapes is fundamental in understanding the dynamic behavior of structures.

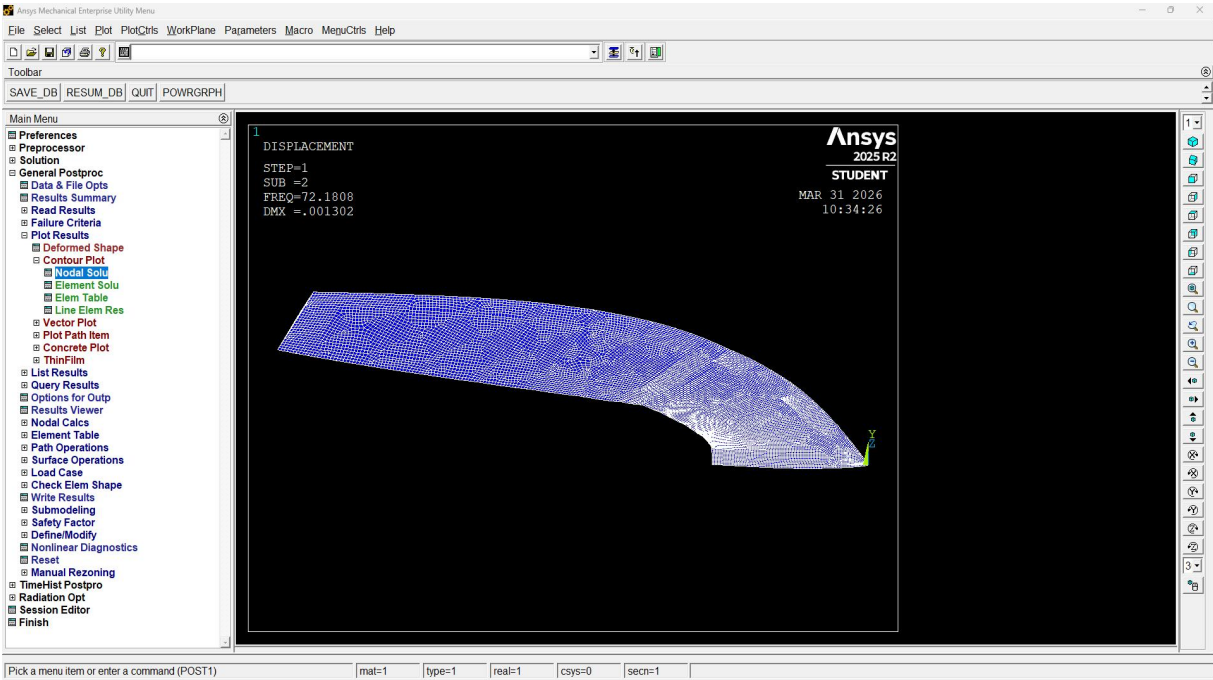
Resonance occurs when the frequency of an external force matches the natural frequency of the structure this leads to a significant increase in vibrational amplitudes, posing a risk of structural instability, material fatigue, and potential damage.

5.3 Visual Results:

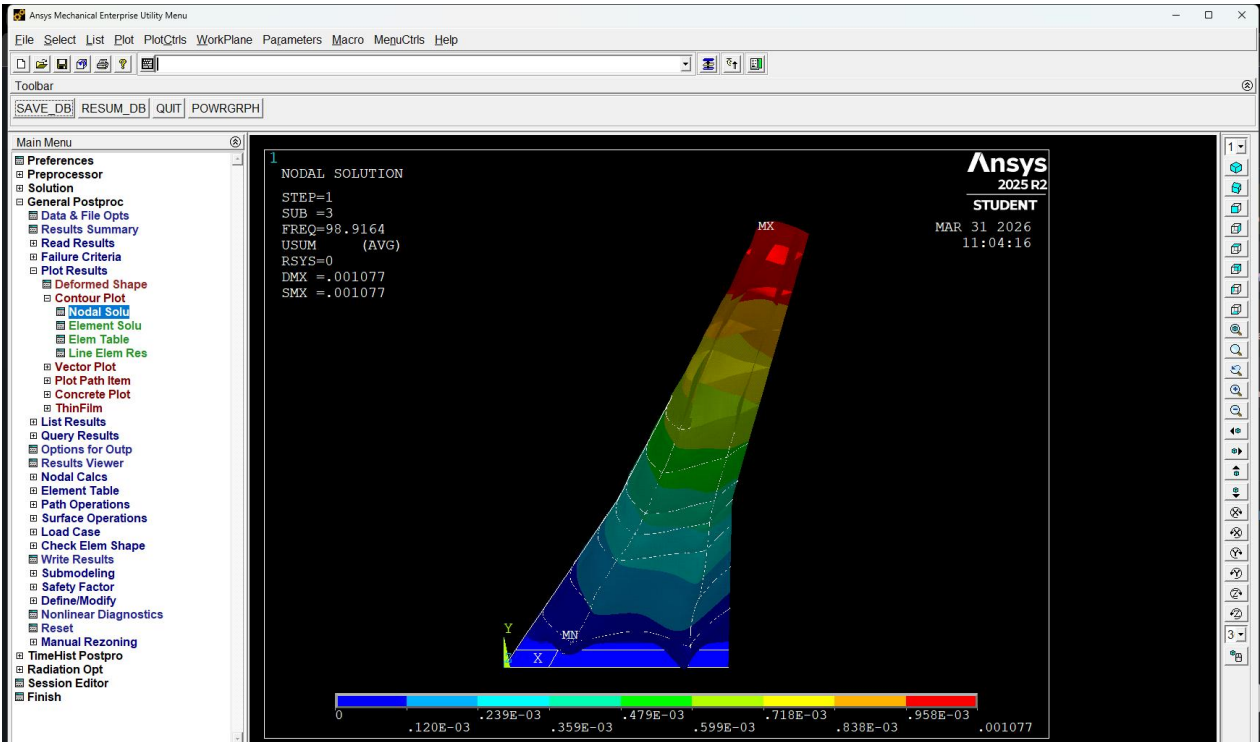
1st Mode Torsion (46.6976 Hz):



1st Mode Bending (72.180 Hz):



2nd Mode Torsion (98.916 Hz):



2nd Mode Bending (132.259 Hz):

